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The Giant Xiarihamu Ni-Co Sulfide Deposit in the East Kunlun Orogenic Belt, Northern Tibet Plateau, China

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Abstract

The Xiarihamu Ni-Co deposit, located in the East Kunlun orogenic belt, northern Tibet plateau, is the secondary largest Ni deposit in China and contains ~157 million metric tons (Mt) sulfide ores with average grades of 0.65 wt % Ni, 0.14 wt % Cu, and 0.013 wt % Co. The intrusion consists of gabbroic and ultramafic portions, the main orebody being hosted in the ultramafic portion. The zircons separated from the gabbronorite and websterite yield SHRIMP U-Pb ages of 405.5 ± 2.7 and 406.1 ± 2.7 Ma, indicating a genetic linkage with the Early Devonian regional magmatism (400–410 Ma). The main orebody includes two large ore pods comprising a pod of disseminated sulfides in the western portion grading into multiple ore sublayers to the east. The ore sublayers consist of net-textured and disseminated sulfides. The distribution and shape of the ore pods and sublayers of the orebody are conformable with the undulating roof and bottom of the ultramafic portion. The unusually high Ni/Cu ratios (~4–18) and extremely low tenors of platinum-group elements (PGE; <4 ppb Ir, <85 ppb Pt, and <115 ppb Pd) of the disseminated sulfides indicate a genetic relationship with low degrees of partial melting of a pyroxenite mantle source. The slightly higher Ir and Ru tenors suggest that the disseminated sulfides in the western portion of the main orebody were segregated from less evolved magma under higher mass ratios of silicate melt/sulfide liquid than the sulfides of the middle and eastern portions.

Introduction

Magmatic sulfide deposits are widely accepted to have formed by concentration of sulfide droplets segregated from mantle-derived mafic or ultramafic magmas. Addition of crustal sulfur and dynamics of magma flow are considered to be the key factors leading to the segregation and concentration of sulfide liquid (e.g., Keays et al., 1981; Keays, 1995; Naldrett, 1998, 2004; Lesher and Keays, 2002; Arndt et al., 2005; Lightfoot and Keays, 2005; Li and Ripley, 2009). Huge amounts of mafic-ultramafic magma produced by high degrees of partial melting of the mantle, probably associated with impact of mantle plumes on the lithosphere, are believed to be necessary for the formation of giant magmatic sulfide deposits, such as the Noril'sk-Talnakh and the Pechenga Ni-Cu-PGE (platinum-group elements) deposits (Russia), the Voisey's Bay Ni-Cu-Co deposit (Canada), the Mt. Keith Ni-Cu deposits (Australia), and the Jinchuan Ni-Cu deposit (China; Ripley et al., 2002; Naldrett, 2004; Beresford et al., 2005; Song et al., 2006, 2008, 2009a; Su et al., 2008; Barnes and Fiorentini, 2012; Lightfoot et al., 2012; Chen et al., 2013, 2015). Thus, the intrusions in large igneous provinces are considered as the best potential targets for magmatic sulfide deposit exploration. However, many Ni-Cu-Co-(PGE) deposits with significant metal reserves have been also discovered in orogenic belts over the world, such as a series of deposits at the southern margins of the Central Asian orogenic belt (e.g., Kalatongke, Huangshan, Huangshandong, and Heishan in northwestern China and Hongqiling in northeastern China; Song and Li, 2009; Song et al., 2011; Qin et al., 2011; Zhang et al., 2011; Xie et al., 2012, 2014; Gao et al., 2013; Wei et al., 2013), the Aguablanca deposit in the Variscan collisional orogeny, Spain (Piña et al., 2008, 2010), the deposits in the Selebi-Phikwe belt, Botswana (Maier et al., 2008), the Santa Rita deposit in the Itabuna-Salvador-Curaca belt, Brazil (Barnes et al., 2013), and the Kotalahti and Vammala belts in Finland (Peltonen, 1995; Makkonen et al., 2008;). Although the Ni-Cu deposits discovered in orogenic belts are on average smaller in sizes and lower in metal grades than those occurring in large igneous provinces, many of them are economically viable or actively mined, such as Santa Rita, Kalatongke, and Huangshandong (Fig. 1). The potential Ni reserves of some deposits in orogenic belts may be more than 1 million metric tons (Mt) of contained metal. An example of such deposits is the Xiarihamu deposit discovered in the East Kunlun orogenic belt, northern Tibet plateau. These discoveries clearly indicate a promising exploration potential for magmatic Ni-Cu deposits in orogenic belts. However, the geologic and geochemical characteristics and effective prospecting criteria for magmatic Ni-Cu deposits in orogenic belts are not well understood. Effects of geometrical features of magma chambers on the physical mechanism of sulfide deposition and geochemical characteristics of Ni-Cu deposits have not been well documented. Geologic and geochemical characteristics allow us to propose that the giant Xiarihamu Ni-Co deposit is the result of concentration of the sulfides segregated from parental magmas originally depleted in PGE and rich in Ni. Zircon SHRIMP U-Pb ages indicate a genetic linkage between the deposit and basaltic magma derived from low-degree partial melting of a pyroxenite mantle source in the Early Devonian.

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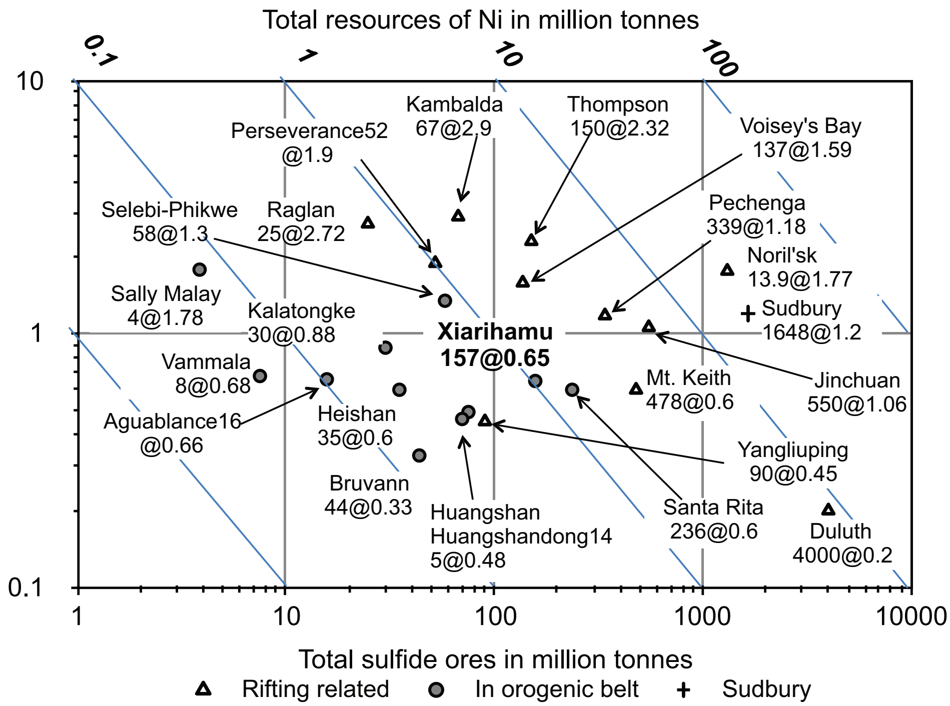

Fig. 1. Plot of Ni grade vs. size of deposit for a range of Ni-Cu-PGE deposits worldwide (Song et al., 2003; Naldrett, 2004; Maier et al., 2008; Mao et al., 2008; Piňa et al., 2008; Song and Lie, 2009; Barnes et al., 2011; Deng et al., 2014; Xie et al., 2014). Million metric tonnes ore at average grade.

Exploration History

Although a few small magmatic sulfide-mineralized intrusions have been discovered on the Tibet plateau since the 1950s, such as Lashuixia, Ni-Cu sulfide deposits were not previously considered as important exploration targets in this large area (Gao et al., 2012; Zhang et al., 2012). The giant Xiarihamu Ni-Co deposit was found by No. 5 Geological and Mineral Survey Institute of Qinghai Bureau of Geology and Mineral Resources (No. 5 GMSI) during a regional exploration program aiming at hydrothermal base and precious metal mineralization in the East Kunlun orogenic belt. The Xiarihamu mafic-ultramafic intrusion was not identified during 1:200000 geologic mapping in the 1970s owing to Quaternary soil cover up to 5 m thick. The Ni mineralization was discovered in 2010 when geologists followed up an outcrop occurrence and stream-sediment Ni-Cr-Co-Cu geochemical anomaly at Xiarihamu, along with exploratory trenches and ground magnetic and electric surveys. An extensive subsequent exploration program supported by the government of Qinghai province included more than 300 diamond drill holes at 40- or 80-m spacing, and defined a resource of ~$1.6*10^8$ t of mainly disseminated sulfide ores with average 0.65 wt % Ni, 0.14 wt % Cu, and 0.013 wt % Co (No. 5 GMSI). This makes the Xiarihamu deposit not only the second largest Ni deposit in China after Jinchuan, but also one of the largest Ni deposits in an orogenic setting in the world (Fig. 1).

Geologic Background

The East Kunlun orogenic belt

The E-W-trending East Kunlun orogenic belt is situated between the Qaidam block to the north and the Songpan-Ganzi block to the south in the northern Tibet plateau, China (Fig. 2). It is bounded by the NE-SW-trending Altyn Tagh fault to the west and the Qinling orogenic belt to the northeast (Xu et al., 2001). The East Kunlun orogenic belt is subdivided into the Northern, Middle and Southern zones by the northern and central Kunlun faults (Fig. 2b).

The Middle Kunlun zone is characterized by widespread Proterozoic metamorphic basement. It comprises the Jinshuikou Group and a number of large Paleozoic and Mesozoic granite plutons (Fig. 2d). The Jinshuikou Group is dominantly composed of gneiss, schist, and marble and was intruded by Neoproterozoic granites (Chen et al., 2006). The Proterozoic metamorphic basement is overlain by Ordovician low-grade metamorphosed clastic rocks. The Ordovician strata are unconformably overlain by terrestrial volcanics of the Early Devonian Maoniushan Formation, which comprises andesite, dacite, and rhyolite with basalt interlayers (Lu et al., 2010). The Maoniushan Formation is uncomformably overlain by Carboniferous and Permian sedimentary and volcanic strata.

Voluminous Paleozoic granitoid plutons were emplaced into the Proterozoic metamorphic rocks and the late Ordovician strata in the middle East Kunlun zone (Mo et al., 2007; Xu et al., 2007; Cui et al., 2011; Fig. 2d). Zircon U-Pb ages (390–420 Ma) indicate that the Paleozoic granitoid plutons occupy ~50% of the area in the eastern portion of the middle East Kunlun zone and were emplaced dominantly in Silurian to Early Devonian (Fig. 2c; Mo et al., 2007; Xu et al., 2007; Cui et al., 2011; Gao and Li, 2011; Liu et al., 2012). In the western portion of the middle East Kunlun zone, the Paleozoic granitoid plutons formed slightly earlier and have been dated at 420 to 440 Ma (Fig. 2c; Lu et al., 2010; Gao and Li, 2011; Liu et al., 2012). Another peak age of granitic magmatism in the

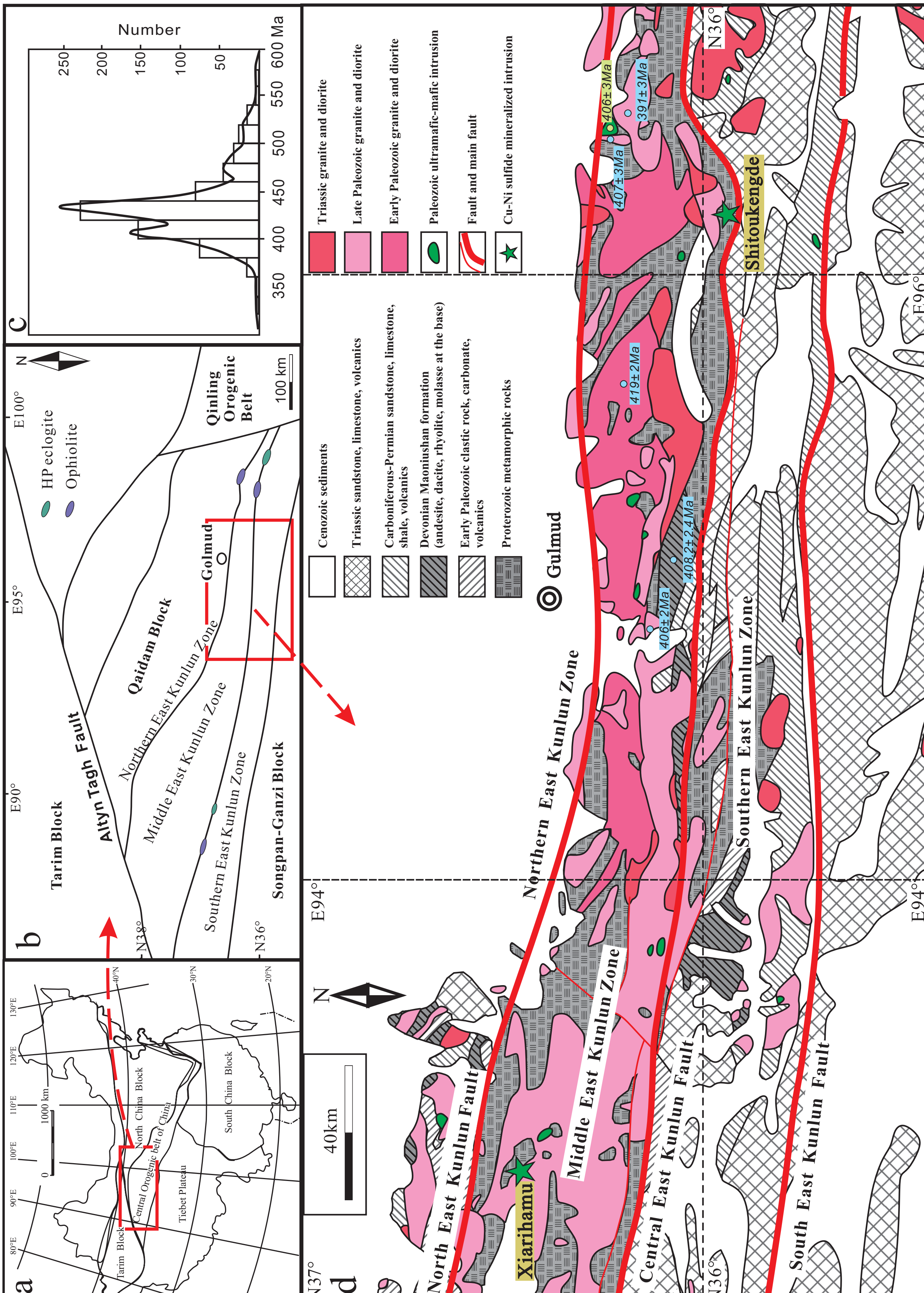

Fig. 2. (a). Tectonic sketch map of China. (b). Simplified tectonic units of the East Kunlun orogenic belt (modified after Xu et al., 2001). (c). Age peaks of the granitoid plutons in the East Kunlun orogenic belt (after Liu et al., 2012). (d). Simplified geologic map of the eastern portion of the East Kunlun orogenic belt (after regional 1:200,000 geologic map).

middle East Kunlun zone is 240 to 260 Ma, consistent with the Triassic volcanism (Huang et al., 2014). Many mafic-ultramafic intrusions occur in the middle East Kunlun zone and a few of them were dated as the Early Devonian age by zircon U-Pb dating using laser ablation-inductively coupled plasma mass spectrometry (LA-ICPMS) or sensitive high resolution ion microprobe (SHRIMP); Fig. 2d). Magmatic sulfide mineralization has also been discovered in another mafic-ultramafic intrusion, Shitoukengde, which is situated ~300 km to the east of the Xiarihamu intrusion at the north side of the Central East Kunlun fault (Fig. 2d).

Similar to the middle East Kunlun zone, the Proterozoic basement in the southern East Kunlun zone also consists of amphibolite, gneiss, schist, and marble (Jiang et al., 1992; Pan et al., 1996; Liu et al., 2005) and is overlain by early Paleozoic weakly metamorphosed clastic sediments, limestone, and volcanic rocks (Jiang et al., 1992; Yang et al., 1996; Bian et al., 2004). The Proterozoic metamorphic rocks and the Devonian strata are restricted to the northern part of the southern East Kunlun zone, whereas the Carboniferous, Permian, and Triassic strata are widespread. Paleozoic and Mesozoic granitic plutons in the southern East Kunlun zone are much smaller than those in the middle East Kunlun zone (Fig. 2d).

Paleozoic ophiolite fragments and high-pressure eclogites occurring along the central East Kunlun fault indicate that the regional faults represent a fossil suture zone between the middle and southern East Kunlun microcontinental blocks (Fig. 2b) (Jiang et al., 1992; Yang et al., 1999, 2000; Chen et al., 2002). The SHRIMP U-Pb ages of the core and rim of the zircon crystals from the coarse-grained eclogite (with means of 934 and 428 Ma, respectively) indicate that the early Neoproterozoic mafic protolith was overprinted by Paleozoic high-pressure metamorphism due to collision (Meng et al., 2013a, b). Ophiolite fragments also occur along the southern East Kunlun fault, which separates the Songpan-Ganzi block from the southern Kunlun zone to the south (Fig. 2b, d; Xiong et al., 2014).

The Xiarihamu intrusion

The Xiarihamu intrusion is emplaced in the Proterozoic Jinshuikou Formation, which dominantly consists of schist, granitic gneiss, and marble. As shown in Figure 3a, to the north of the Xiarihamu intrusion, an E-W-trending syenite pluton with high 10,000 × Ga/Al ratios of 3.77 ~ 4.10, dated at 391 ± 1.4 Ma, was emplaced in the Jinshuikou Formation (Wang et al., 2013); a much larger undated granitic pluton occurs to the south of the Xiarihamu intrusion. The Early Devonian Maoniushan Formation unconformably overlying Ordovician marine sedimentary sequence appears to the north of Xiarihamu and mainly consists of andesite and basaltic andesite (Fig. 3a).

Other two small mafic-ultramafic intrusions, HS27 and HS28, with lengths less than 1,000 m and widths of 80 to 500 m, are located to the south of the Xiarihamu intrusion and emplaced in the Jinshuikou Formation (Fig. 3a). Both of them are E-W trending and have similar rock assemblages with the Xiarihamu intrusion, consisting mainly of gabbro and pyroxenite with very weak sulfide mineralization. LA-ICPMS

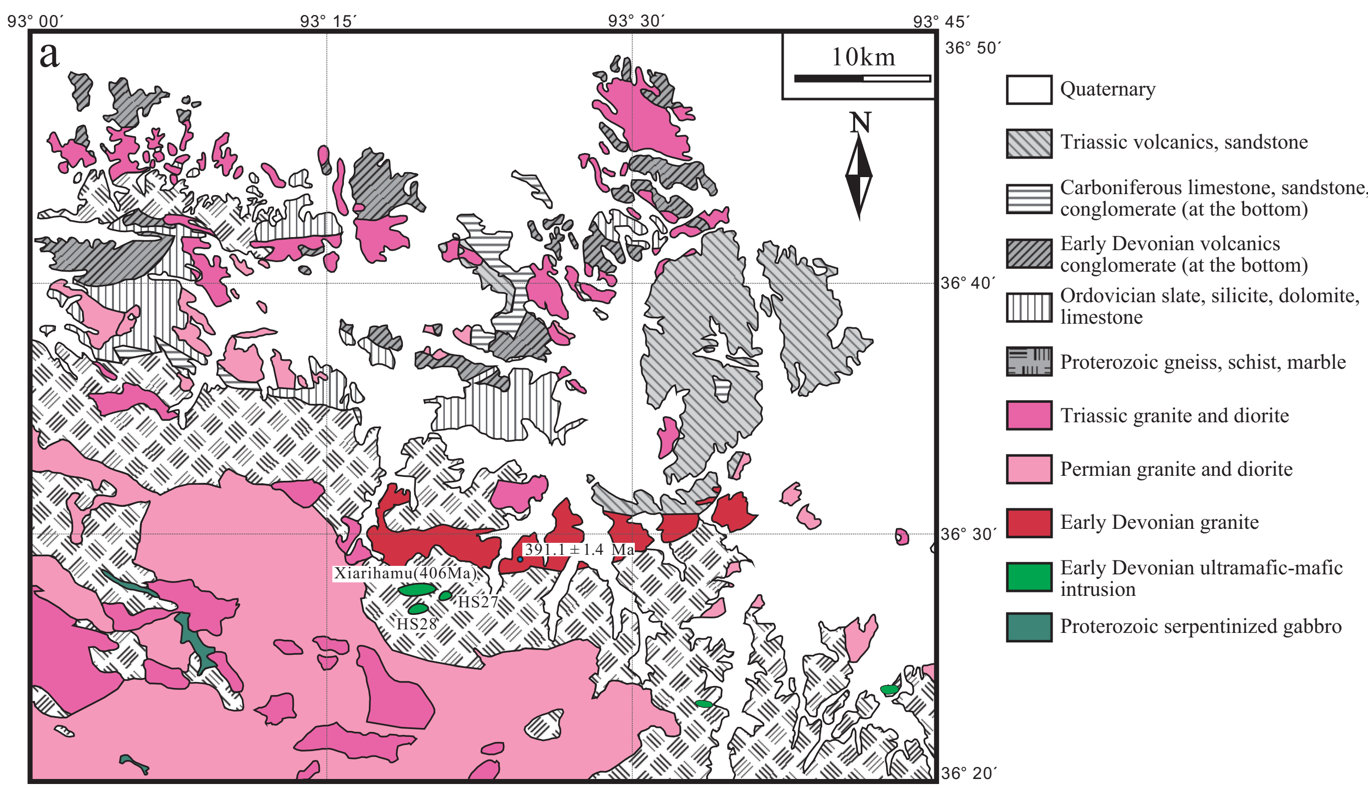

Fig. 3. (a). Simplified geologic map of the Xiarihamu area (after regional 1:200,000 geologic map). (b). Geologic map at Xiarihamu (modified after No. 5 GMSI, 2013). (c). Thicknesses of different rock types of the ultramafic portion of the Xiarihamu intrusion; types of the footwall and hanging-wall rocks are also shown.

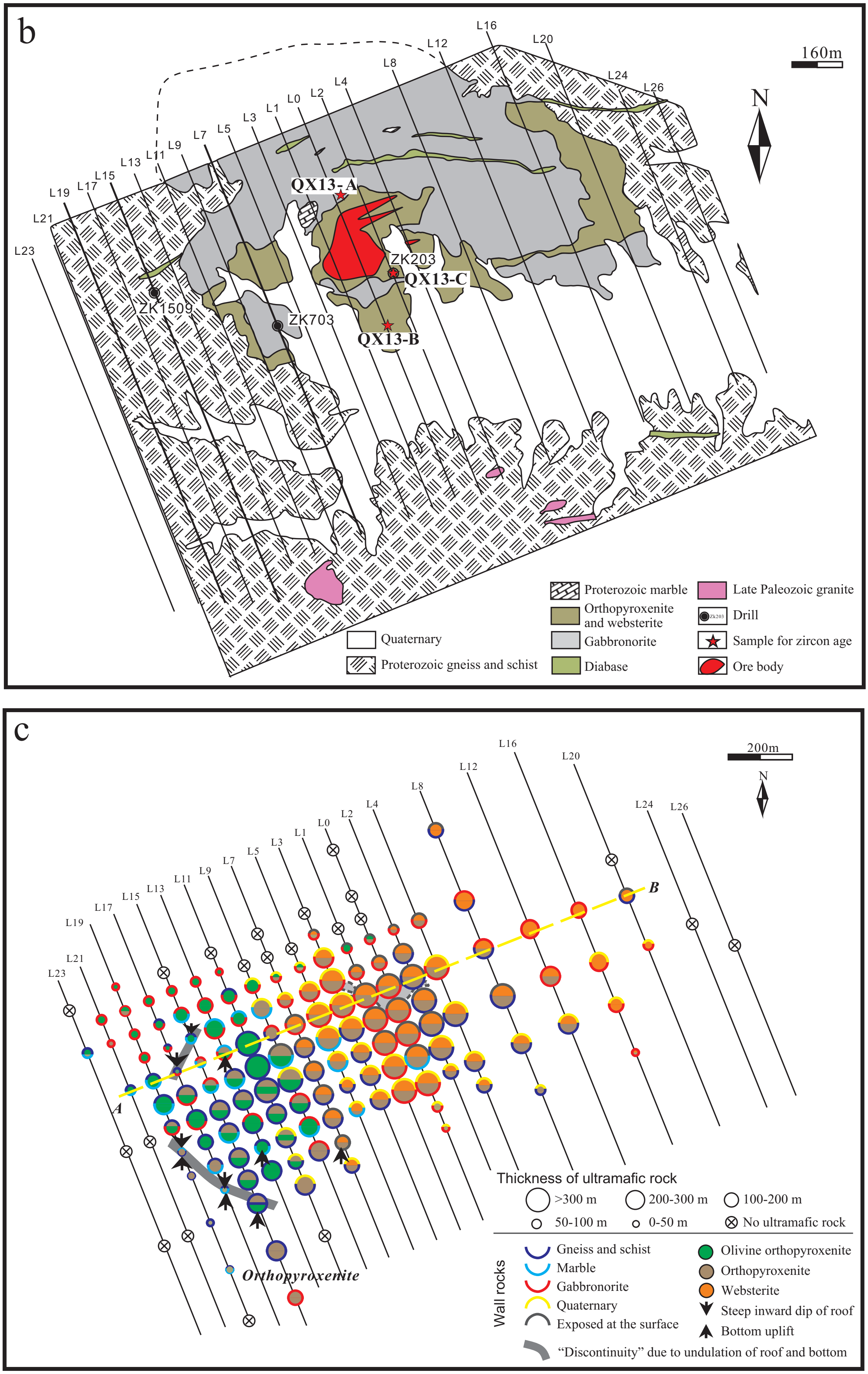
b
160m
N
QX13-A
ZK203
QX13-C
ZK1509
ZK703
QX13-B
Proterozoic marble
Orthopyroxenite and websterite
Gabbronorite
Diabase
Late Paleozoic granite
Drill
Sample for zircon age
Ore body
Quaternary
Proterozoic gneiss and schist
c
200m
N
B
A
Orthopyroxenite
Thickness of ultramafic rock
>300 m
200-300 m
100-200 m
50-100 m
0-50 m
No ultramafic rock
Wall rocks
Gneiss and schist
Marble
Gabbronorite
Quaternary
Exposed at the surface
Olivine orthopyroxenite
Orthopyroxenite
Websterite
Steep inward dip of roof
Bottom uplift
"Discontinuity" due to undulation of roof and bottom

Fig. 3. *(Cont.)*

zircon U-Pb dating indicated that the HS27 intrusion formed at 422.4 ± 1.0 Ma, earlier than the Xiarihamu intrusion (No. 5 GMSI). The following descriptions complement those recently published by Li et al. (2015).

Petrography and Geometry of the Xiarihamu Intrusion

As shown by the simplified geologic map modified in 2013 and 2014 (Fig. 3b), the Xiarihamu intrusion is irregularly shaped with a length of ~1,400 m and width of ~900 m. The western part of the intrusion is not exposed and the southern part is covered by Quaternary clastic sediments. Drill holes reveal that the Xirihamu intrusion is a NE-SW-extending oval-shaped body with a length of ~2,000 m and width of more than 1,400 m (Fig. 3b). The intrusion is entirely emplaced in the Proterozoic Jinshuikou Formation and is dominantly in contact with granitic gneiss, schist, and marble. The metamorphic country rocks are all sulfide poor.

The intrusion is composed of a gabbroic portion and ultramafic portion (Fig. 3b). Although the Jinshuikou metamorphic rocks are the common country rocks of the ultramafic portion, gabbroic rocks form the footwall and hanging wall at some locations, particularly to the east (Figs. 3c, 4b). This illustrates that the upper part of the gabbroic portion has been eroded while the ultramafic portion is well preserved.

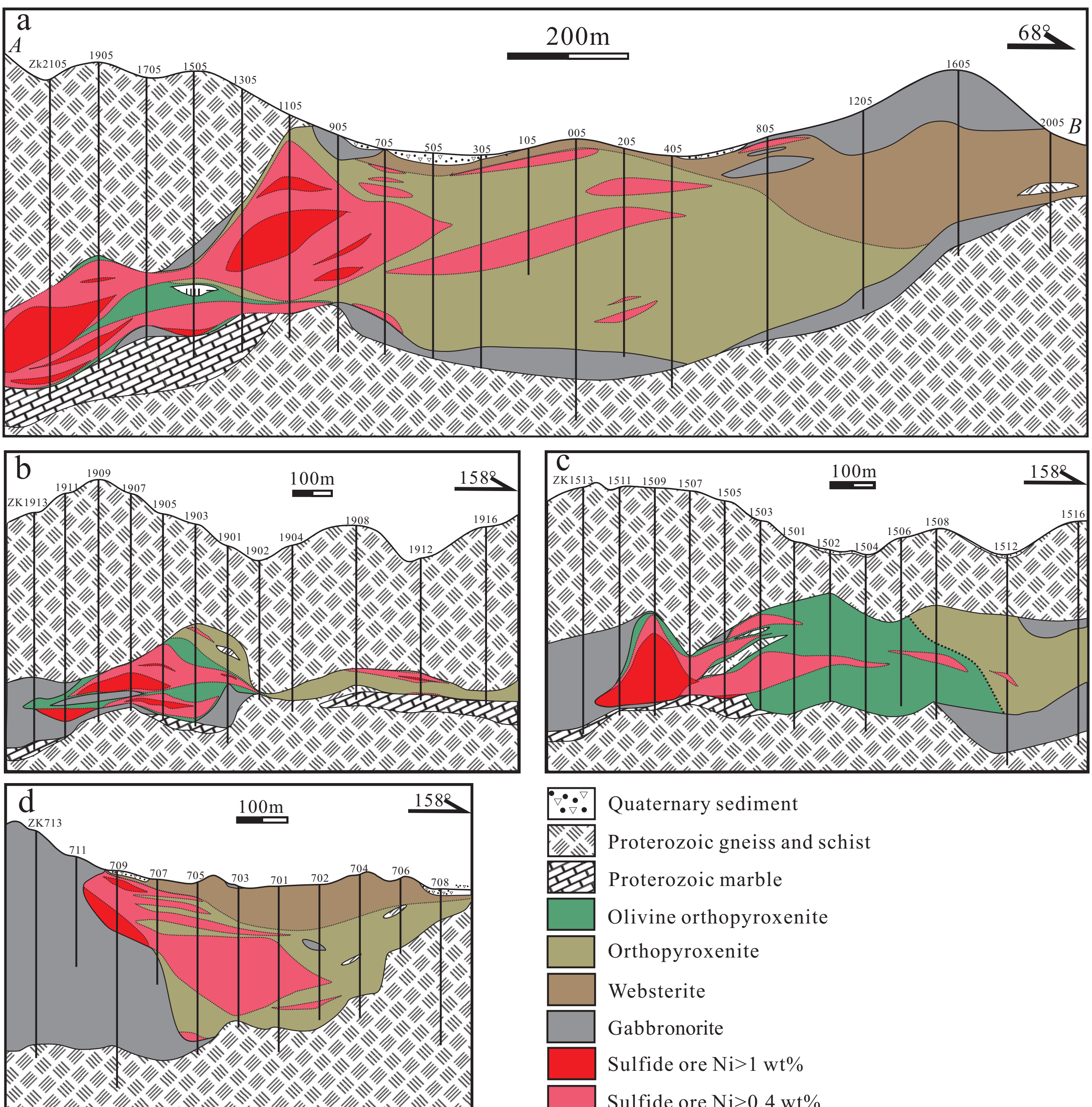

Fig. 4. SW-NE-trending cross section A-B across the northern part of the Xiarihamu ultramafic portion (a) and NW-SE-trending exploration lines 19 (b), 15 (c), and 7 (d). Locations of the cross sections are shown in Figure 3c.

Observations of both surface outcrop and drill holes have confirmed no chilled margins and no mineral-size variation between the gabbroic and the ultramafic portions. However, gabbroic blocks enclosed within the ultramafic portion appear in many drill holes (Fig. 4). Thus, it is reasonable to deduce that the ultramafic portion intruded slightly later but before complete solidification of the gabbroic portion. This is consistent with the concordant zircon U-Pb age data of the gabbronorite and websterite within error range (Fig. 8). A small gabbroic body occurs to the south of the ultramafic portion (Li et al., 2015).

Petrography

The gabbroic portion composes of medium-grained gabbronorite with minor lithologic variation. The gabbronorite contains 50 to 60 modal % plagioclase, 25 to 35% orthopyroxene, 10 to 20% clinopyroxene, interstitial hornblende and biotite (<5%), and minor magnetite and sulfide (Fig. 5a). Tabular plagioclase crystals are in some cases poikilitically enclosed in pyroxene.

The ultramafic portion principally comprises medium- and fine-grained rocks, olivine orthopyroxenite overlain by orthopyroxenite in the west and orthopyroxenite overlain by websterite in the east, approximately on the two sides of the exploration line 9 (Fig. 4). The olivine orthopyroxenite consists of 10 to 40% olivine, 50 to 80% orthopyroxene, <10% clinopyroxene and plagioclase, minor hornblende, and biotite. Orthopyroxene occurs as large oikocrysts up to 1 cm across with abundant olivine chadacrysts or as granular or interstitial grains. Olivine chadacrysts within the orthopyroxene show resorbtion due to reaction between olivine and magma and are considerable finer grained than those outside. All the other silicate minerals in the olivine orthopyroxenite, including clinopyroxene, plagioclase, hornblende, and biotite, are interstitial between olivine and orthopyroxene (Fig. 5b). The orthopyroxenite contains 60 to 90% orthopyroxene, <10% olivine, <10% plagioclase, <10% clinopyroxene, and minor hornblende and biotite (Fig. 5c).

To the east of the exploration line 9, the olivine orthopyroxenite wedges out and the orthopyroxenite becomes dominant and overlain by websterite as clinopyroxene increasing upward in the ultramafic portion. The websterite contains 10 to 20% clinopyroxene and ~80% orthopyroxene and <10% plagioclase, hornblende, and biotite (Fig. 5d). Plagioclase occurs as oikocrysts locally in excess of 5 modal % in the websterite. Thin layers of gabbronorite occur at the contact zone with the Jinshuikou metamorphic rocks in some locations and contain ~40% plagioclase, 30 to 40% orthopyroxene, 10 to 20% clinopyroxene, and less than 5% hornblende and biotite.

Geometry of the ultramafic portion

The ultramafic portion forms an irregular rectangle in surface projection with a length of ~1,700 m and a width of ~1,200 m (Fig. 3c). The ultramafic portion dips shallowly to the west

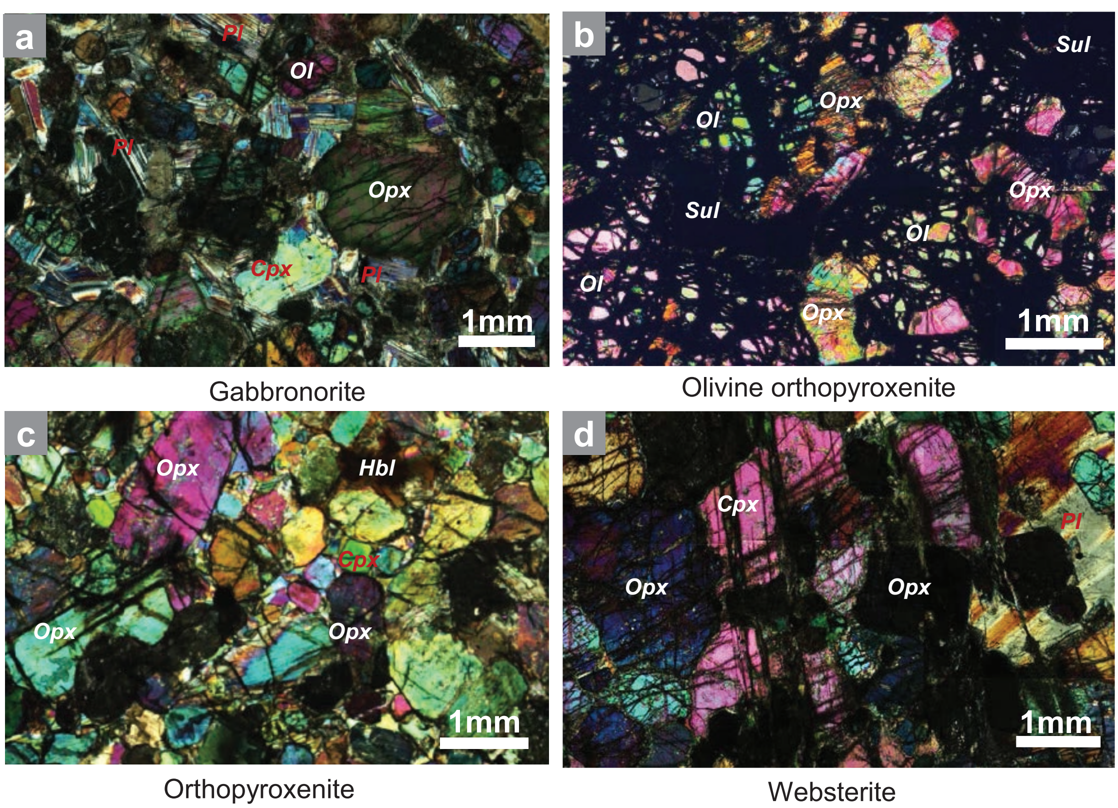

Fig. 5. Microscope photos under cross-polarized light. (a) Fine-grained gabbro comprising granular pyroxene, plagioclase and minor olivine. (b). Fine- to medium-grained olivine orthopyroxenite comprising poikilitic orthopyroxene and olivine grains and minor interstitial sulfides. (c). Fine-grained orthopyroxenite comprising granular and poikilitic pyroxene and minor interstitial hornblende. (d). Medium-grained websterite comprising poikilitic orthopyroxene hosting minor fine olivine grains. Cpx = clinopyroxene, Hbl = hornblende, Ol = olivine, Opx = orthopyroxene, Sul = sulfide.

and its base extends downward to ~500-m depth (Fig. 4). The undulating top and bottom contacts of the ultramafic portion give rise to the large variation in thicknesses, particularly in the western part (Figs. 3c, 4a).

In the western part of the ultramafic portion, a SE-NW-extending undulating footwall contact with Jinshuikou metamorphic rocks extends from the exploration line 19 to line 15 (Figs. 3c, 4b, c). The undulating footwall contact and steep inward dip of the hanging wall form a SE-NW-trending "discontinuity" as narrow as 10 to 30 m between exploration lines 19 and 17 (Fig. 4b). This discontinuity was formed during magma emplacement. To the south of the discontinuity, the ultramafic portion becomes much thinner (<50 m thick; Fig. 3c) and the olivine contents in the olivine orthopyroxenite decrease. Another short NE-trending discontinuity in the ultramafic portion appears between exploration lines 17 and 15 due to an inward dip of the roof (Figs. 3c, 4c). The ultramafic portion becomes markedly thicker to the east of the two discontinuities.

Toward the exploration line 13, the ultramafic portion becomes thicker as the roof rises and the base dips to the east. The ultramafic portion becomes more than 300 m thick between exploration lines 13 and 8. The thickest part of the ultramafic portion is developed between exploration lines 7 and 8, where the orthopyroxenite is overlain by plagioclase websterite (Figs. 3c, 4a, d). To the east of exploration line 8, a topographic high in the base is associated with thinning of the ultramafic section (Figs. 3c, 4a).

The Magmatic Sulfide Mineralization

Most of the sulfide ores are hosted in the ultramafic portion at Xiarihamu; the gabbroic portion only contains weak mineralization in a few locations. More than 95% of the Ni reserves are hosted in the largest orebody, M1 (abbreviated as M1 below), whereas the other small orebodies contain only 5% of the reserves (No. 5 GMSI). The main body of M1 is continuous and hosted in both olivine orthopyroxenite and orthopyroxenite to the west of line 9 (Figs. 4, 6), whereas to the east

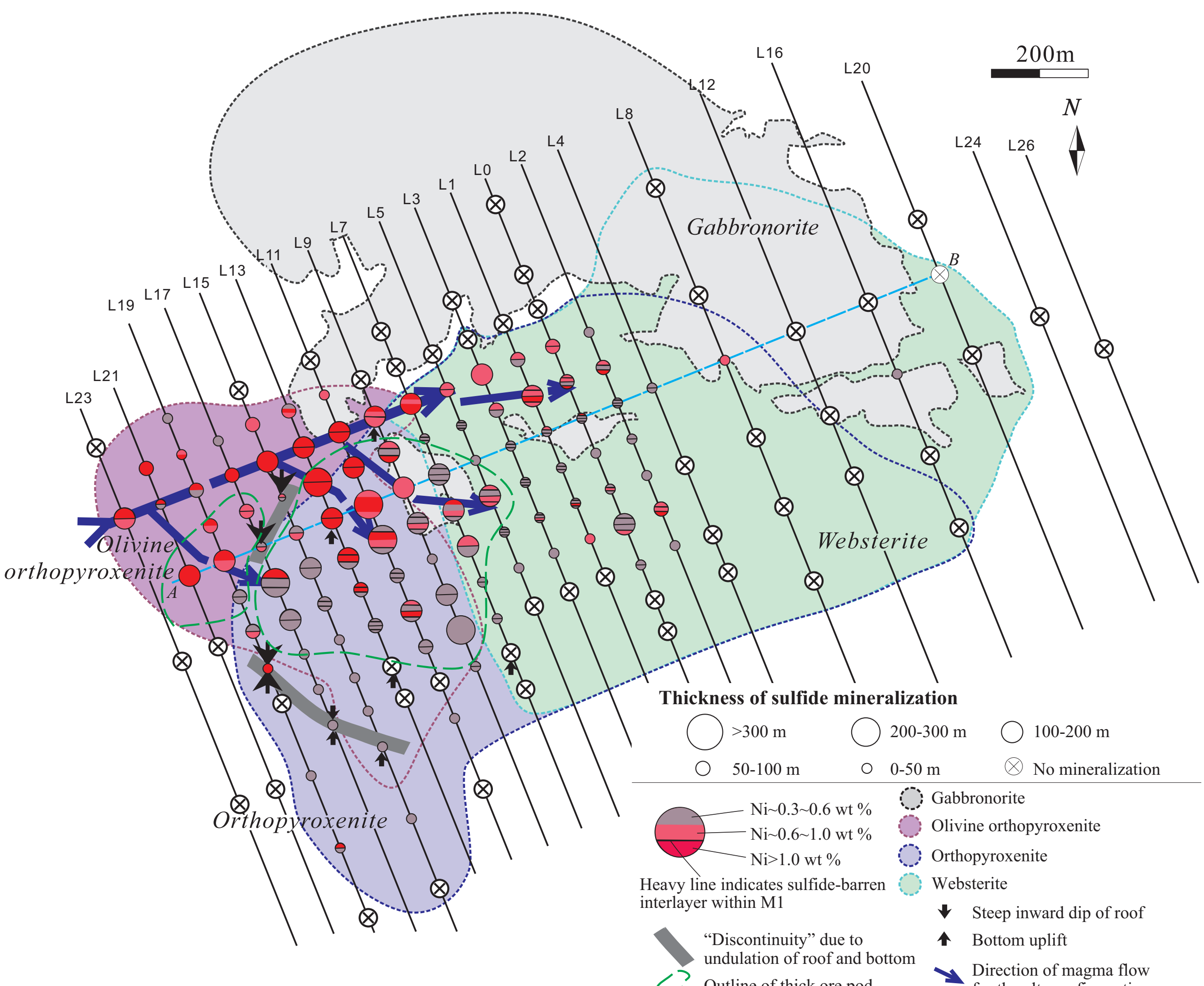

Fig. 6. Thickness variation of the main orebody M1 and different Ni grades of the sulfide ores at Xiarihamu. The thick ore pods are located on the two sides of the discontinuities in the western part of the ultramafic portion. The sl,urries caring sulfide droplets are inferred to have flowed from the west toward to east and southeast. Lines lavbeled L23, L21 etc. refer to the "exploration lines" mentioned in the text. There are more sulfide-barren interlayers within the main orebody M1 to the east.

of line 7 it is divided into several ore sublayers and hosted only by the orthopyroxenite. The geologic features of M1, including its shape and thickness, ore textures, and grades, are distinguished on the two sides of line 5.

To the west of line 5, M1 has two thick ore pods on the two sides of the NE-SW-trending discontinuity between lines 15 and 19 (Figs. 4b, 6). Pinching and swelling of the ore pods in three dimensions are conformable with swelling of the ultramafic portion. The eastern ore pod is larger than the western ore pod (Figs. 3c, 4a, 6). A few lens-shaped blocks of the Jinshuikou metamorphic rock and gabbronorite are enclosed in the western ore pod (Fig. 4b). To the south of the NW-SE-trending discontinuity between lines 17 and 19, the orebody becomes much thinner (<50 m thick) and usually contains less than 15% sulfides; a sulfide-barren orthopyroxenite layer, up to 100 m in thickness, occurs beneath the sulfide ore layer (Fig. 4b, c). The eastern ore pod between lines 5 and 17 is up to 300 m thick and 600 m wide where the roof of the ultramafic portion rises to the east. The two ore pods are dominantly composed of disseminated ores, which contain less than 20 modal % sulfides (Fig. 7a-b). Sulfide contents of the M1 orebody tend to decrease from northwest to southeast in both orebody expansions (Fig. 6).

M1 tends to be divided into sublayers to the south and east as a result of intercalated layers of weakly mineralized or sulfide-barren orthopyroxenite within the orebody, particularly to the east of line 7 (Figs. 4d, 6). Additionally, many thin, sulfide-barren orthopyroxenite bands less than 1 m thick occur within the ore sublayers. A layer of sulfide-barren orthopyroxenite with variable thickness beneath the lowest disseminated ore sublayer occurs to the east of line 7 (Figs. 4, 6). Each ore sublayer is commonly less than 50 m thick and those at the lower level have lower Ni and Cu grades than those in the upper level (No. 5 GMSI). The ore types include disseminated ore (sulfides = 5–20 modal %), net-textured or blotchy textured ores (sulfides = 20–70%) as well as massive ore (sulfides >75%). In the blotchy textured ores, oblate patches containing as high as 50 to 70% sulfides and orthopyroxene grains within a contiguous sulfide matrix are separated by sulfide-free orthopyroxenite (Fig. 7c-d). The oblate patches are variable in size (commonly less than $3 \times 2 \times 2$ cm^3) and shape and have sharp or blurry boundaries. Variable-sized bean-shaped bodies of massive sulfides, with common thickness of a few meters (up to 15 m) and length of ~30 m (up to 80 m) and width of ~20 m (less than 40 m), have been found in different levels within the net-textured and blotchy textured ores (Fig. 7c). Although the sulfide ores between the exploration lines 7 to 2 have higher sulfide contents than those to the west of the exploration line 9, their Ni grades are commonly lower than 1.0 wt % (Fig. 6). The small dip angles of the orebody M1 to southwest (<15°) indicate that original orientation of the Xiarihamu intrusion was preserved during the later stages of the orogeny (Fig. 4).

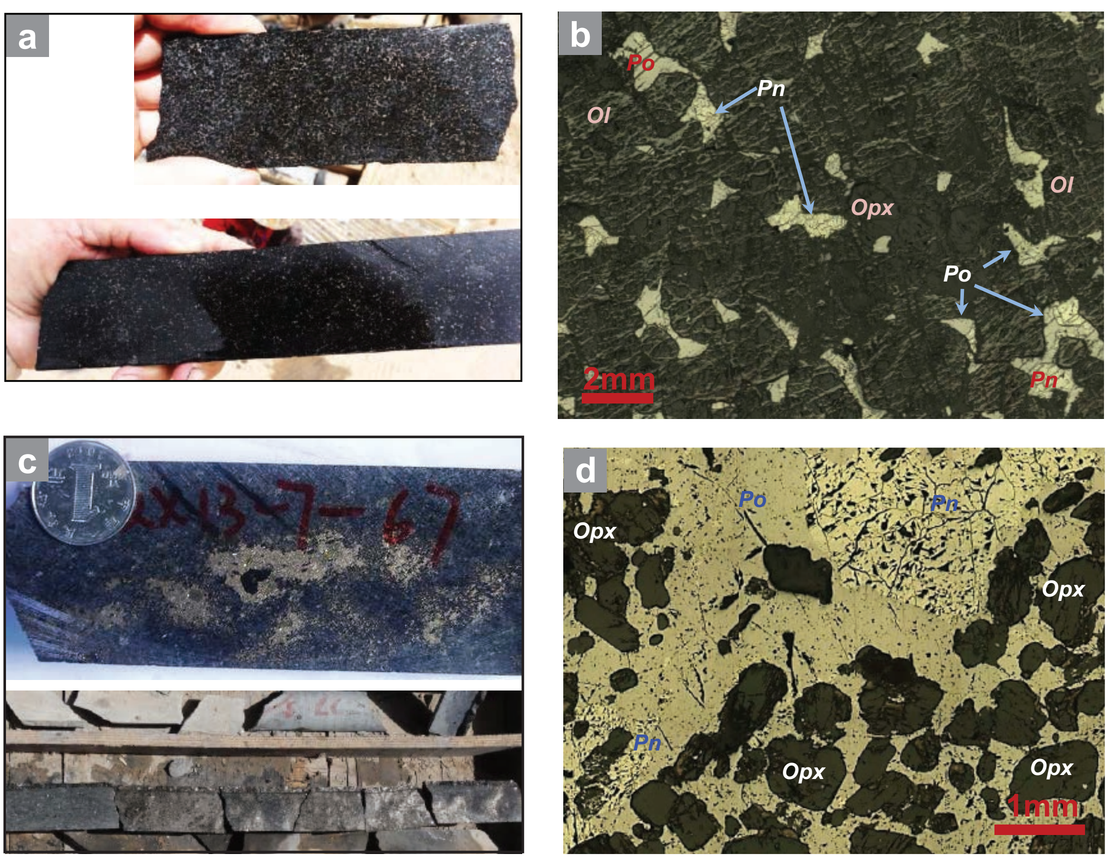

Fig. 7. (a). and (b). Disseminated sulfide ores hosted in the olivine orthopyroxenite from drill core of ZK1905; sulfides occur interstitially between olivine and pyroxene crystals. (c). and (d). Net-textured and blotchy textured sulfide ores in middle and eastern part of M1 (ZK703), silicates are enclosed by the sulfides. Ol = olivine, Opx = orthopyroxenite, Pn = pentlandite, Po = pyrrhotite.

The sulfide ores have a typical magmatic sulfide assemblage, including pyrrhotite, pentlandite, and subordinate but ubiquitous chalcopyrite and cubanite (Fig. 7). Valeriite, covellite, violarite, and gersdorffite are minor components. Serpentine, talc, tremolite, chlorite, and magnetite are common in the sulfide ores as products of the alteration of silicates (Fig. 7).

Analytical Methodology

Zircon U-Pb dating

Sixty-kg samples of one gabbronorite (QX13-A) and two websterites (QX13-B from the surface and QX13-C from the bore hole ZK203, respectively; Fig. 3b) were crushed for zircon crystal separation. Following conventional magnetic and heavy liquid separation, 60 to 100 zircon crystals from each sample were handpicked under a binocular microscope for U-Pb dating. The zircon grains in size range 20 to 80 μm were mounted in epoxy and polished before cathodoluminescence (CL) observation using a HITACHI S3000-N scanning electron microscope. Grains larger than 40 μm in diameter were selected for in situ U-Th-Pb analyses by SHRIMP II at Beijing SHRIMP Center, Institute of Geology, Chinese Academy of Geological Sciences. Analytical procedures follow those of Compston et al. (1992), Williams (1998), and Song et al. (2002). The standard zircons TEM (416.8 Ma, with variable U content) (Black et al., 2003) were used to correct for interelement fractionation, and U, Th, and Pb concentrations were determined based on the standard zircons M257 (561.3 Ma, U = 840 ppm; Nasdala et al., 2008). A primary ion beam of 25-μm diam, with a current of 3.0 nA was used. Five data collection cycles were performed for each analysis. Data processing was carried out using the SQUID 1.03 and Isoplot/Ex2.49 programs of Ludwig (2001a, b), and the ^{204}Pb-based method of common Pb correction was applied. Uncertainties for each analysis are at 1σ, whereas the weighted mean age is quoted at 2σ (Table 1). The decay constants recommended by Steiger and Jager (1977) were used for age calculations and the model proposed by Stacey and Kramers (1975) was used in calculations of initial common Pb, using measured ^{204}Pb/^{206}Pb and contemporaneous common Pb isotope compositions.

Measurements of Ni, Cu, S and PGE

Samples were collected from drill cores ZK1509, ZK703, and ZK203 to constrain the compositional variations of the orebody M1 in the western and central parts of the ultramafic portion (Fig. 3b). Entire samples were crushed with steel jaws to –10 mesh, and then ~200 g of this fraction were ground to –200 mesh powder using a tungsten carbide ring mill.

Whole-rock Ni and Cu contents of samples containing more than 2 modal % sulfides were measured using VarianICP735-ES inductively coupled plasma emission spectrometer and Perkin Elmer Elan 9000 inductively coupled plasma mass spectrometer (ICP-MS), respectively, at the ALS Chemex (Guangzhou) Co. Ltd. Whole-rock S contents were measured using a gravimetric method and IR absorption with the detection limits ~0.01 wt %. The analytical precisions are ~8% of amount present for S, ~3% for Ni and Cu. Platinum-group elements were determined by isotope dilution (ID)-ICP-MS using an improved digestion technique using sealed beakers sealed in a stainless steel pressure bomb (Qi et al., 2011).

Table 1. SHRIMP U-Pb Isotope Data for Zircons of the Samples from the Xiarihamu Intrusion

Sample/spot	^{206}Pb[1] (%)	^{206}Pb[2] (ppm)	U (ppm)	Th (ppm)	^{232}Th/^{238}U	^{207}Pb/^{206}Pb	±1σ (%)	^{207}Pb/^{235}U	±1σ (%)	^{206}Pb/^{238}U	±1σ (%)	Apparent ages (Ma) ^{206}Pb/^{238}U	±1σ	^{207}Pb/^{206}Pb	±1σ	^{208}Pb/^{232}Th	±1σ
QX13-A Gabbronorite																	
QX13-A-1.1	0.29	25.7	454	264	0.60	0.0538	2.9	0.4870	3.3	0.0656	1.6	409.7	6.3	362	66	401.0	11.0
QX13-A-2.1	0.00	55.6	991	549	0.57	0.0537	1.7	0.4830	2.4	0.0653	1.7	407.5	6.6	356	39	392.2	7.4
QX13-A-3.1	0.02	165.0	2942	2407	0.85	0.0541	0.6	0.4886	1.6	0.0655	1.5	408.8	5.9	377	12	386.2	5.9
QX13-A-4.1	0.04	43.5	773	568	0.76	0.0539	1.5	0.4870	2.2	0.0655	1.5	409.0	6.1	369	35	387.5	7.2
QX13-A-7.1	0.07	31.9	573	256	0.46	0.0538	1.6	0.4800	2.2	0.0647	1.6	404.0	6.1	363	36	389.4	8.6
QX13-A-8.1	0.06	29.6	513	407	0.82	0.0538	1.5	0.4980	2.1	0.0671	1.5	418.9	6.3	363	34	396.1	7.5
QX13-A-9.1	4.78	49.8	833	586	0.73	0.0553	7.5	0.5050	7.6	0.0662	1.6	413.4	6.3	422	170	393.0	24.0
QX13-A-10.1	0.00	47.0	834	699	0.87	0.0536	1.1	0.4851	1.8	0.0657	1.5	409.9	6.0	354	24	381.2	8.0
QX13-A-11.1	0.00	80.5	1456	969	0.69	0.0544	1.0	0.4831	1.8	0.0644	1.5	402.1	6.0	389	21	383.9	6.5
QX13-A-12.1	0.06	39.8	727	463	0.66	0.0546	1.6	0.4800	2.2	0.0637	1.5	398.3	5.9	395	36	390.1	7.6
QX13-A-13.1	--	85.4	1565	1173	0.77	0.0551	0.8	0.4823	1.8	0.0635	1.7	397.0	6.4	415	17	381.9	6.8
QX13-A-14.1	0.00	63.2	1136	634	0.58	0.0544	0.9	0.4861	1.8	0.0648	1.5	404.8	5.9	388	21	375.2	9.3
QX13-A-18.1	0.06	87.2	1575	1116	0.73	0.0540	0.9	0.4798	1.7	0.0644	1.5	402.3	5.8	373	20	377.4	6.2
QX13-A-19.1	0.03	44.6	786	623	0.82	0.0547	1.5	0.4980	2.1	0.0660	1.5	412.3	6.1	398	33	389.3	7.1
QX13-A-20.1	0.04	38.1	682	613	0.93	0.0537	1.4	0.4820	2.2	0.0651	1.7	406.3	6.6	358	32	377.5	7.3
QX13-A-21.1	0.00	154.0	2734	2411	0.91	0.0542	0.6	0.4910	1.6	0.0657	1.5	409.9	5.9	381	13	381.7	6.3
QX13-A-25.1	0.03	33.3	602	277	0.48	0.0552	1.7	0.4910	2.3	0.0645	1.5	402.9	6.0	421	37	384.9	8.8
QX13-A-26.1	--	90.6	1660	730	0.45	0.0554	0.8	0.4850	1.7	0.0635	1.6	397.1	6.1	427	17	380.1	6.6
QX13-A-27.1	0.04	40.3	715	322	0.47	0.0542	1.3	0.4896	2	0.0655	1.5	408.9	6.1	380	30	392.0	7.9
QX13-A-28.1	0.00	102.0	1874	1249	0.69	0.0543	0.7	0.4744	1.7	0.0634	1.6	396.1	6.1	383	16	370.9	6.3
QX13-A-32.1	0.00	46.0	839	546	0.67	0.0538	1.1	0.4733	1.9	0.0638	1.5	398.9	5.8	362	24	370.5	6.5

Table 1. *(Cont.)*

Sample/spot	^{206}Pb[1] (%)	^{206}Pb[2] (ppm)	U (ppm)	Th (ppm)	$^{232}Th/^{238}U$	$^{207}Pb/^{206}Pb$	±1σ (%)	$^{207}Pb/^{235}U$	±1σ (%)	$^{206}Pb/^{238}U$	±1σ (%)	Apparent ages (Ma) $^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{232}Th$	±1σ
QX13-B Websterite																	
QX13-B-2.1	0.05	23.4	411	323	0.81	0.0538	1.6	0.4910	2.3	0.0663	1.7	413.6	6.9	361	35	407.5	8.4
QX13-B-4.1	0.09	69.1	1207	934	0.80	0.0545	1.6	0.5000	2.2	0.0665	1.5	415.1	6.0	393	37	393.8	7.2
QX13-B-5.1	--	21.4	373	180	0.50	0.0557	1.8	0.5140	2.3	0.0670	1.6	418.1	6.3	438	39	413.4	9.5
QX13-B-10.1	0.00	37.3	672	523	0.80	0.0544	1.1	0.4844	2	0.0646	1.6	403.5	6.2	387	26	391.7	7.1
QX13-B-11.1	0.04	84.8	1518	1153	0.78	0.0544	0.9	0.4875	1.7	0.0650	1.5	405.9	5.9	388	19	383.3	6.4
QX13-B-12.1	0.06	29.1	517	393	0.79	0.0544	1.4	0.4920	2.2	0.0656	1.6	409.8	6.3	387	33	395.7	7.6
QX13-B-13.1	--	26.0	458	216	0.49	0.0555	1.4	0.5060	2.1	0.0662	1.5	413.0	6.2	433	30	422.0	8.6
QX13-B-16.1	0.01	69.2	1223	994	0.84	0.0546	0.9	0.4956	1.7	0.0658	1.5	411.1	6.0	395	20	394.9	6.5
QX13-B-18.1	0.01	37.0	664	409	0.64	0.0539	1.2	0.4824	2	0.0649	1.6	405.6	6.3	366	28	390.3	7.4
QX13-B-20.1	0.02	73.2	1318	1259	0.99	0.0546	0.9	0.4864	1.8	0.0646	1.5	403.8	5.8	395	21	378.6	6.1
QX13-B-22.1	0.00	40.9	724	561	0.80	0.0543	1.1	0.4921	1.9	0.0658	1.5	410.5	6.0	383	24	384.3	8.3
QX13-B-23.1	0.15	92.4	1670	1443	0.89	0.0542	1.0	0.4808	1.8	0.0644	1.5	402.0	5.8	379	23	377.0	6.2
QX13-B-27.1	--	29.0	518	239	0.48	0.0548	1.3	0.4930	2	0.0652	1.5	407.5	6.1	404	30	407.3	8.2
QX13-B-28.1	--	20.9	373	185	0.51	0.0553	1.5	0.4980	2.2	0.0654	1.6	408.2	6.2	423	33	393.5	8.4
QX13-B-29.1	--	54.8	982	673	0.71	0.0552	0.9	0.4942	1.8	0.0649	1.5	405.4	5.9	421	21	385.0	7.4
QX13-B-32.1	0.00	69.9	1255	835	0.69	0.0542	0.8	0.4842	1.7	0.0648	1.5	404.8	5.8	379	19	381.8	6.4
QX13-B-33.1	--	38.9	693	564	0.84	0.0552	1.1	0.4973	1.9	0.0653	1.5	407.9	6.0	421	25	394.4	6.9
QX13-B-36.1	--	68.9	1240	826	0.69	0.0543	1.1	0.4836	1.9	0.0646	1.6	403.7	6.2	382	24	376.9	6.5
QX13-C Websterite																	
QX13-C-3.1	0.03	36.8	664	519	0.81	0.0547	1.9	0.4860	2.4	0.0645	1.5	403.0	6.0	399	42	384.4	7.1
QX13-C-4.1	0.10	21.1	378	194	0.53	0.0543	2.2	0.4880	2.7	0.0651	1.6	406.5	6.2	386	49	396.0	9.5
QX13-C-6.1	0.00	30.8	564	425	0.78	0.0547	1.3	0.4803	2	0.0636	1.5	397.7	5.9	401	29	384.0	7.0
QX13-C-7.1	--	24.8	438	291	0.69	0.0555	1.4	0.5060	2.1	0.0661	1.6	412.3	6.2	434	32	414.5	8.2
QX13-C-18.1	--	17.5	308	148	0.50	0.0559	1.7	0.5110	2.4	0.0663	1.6	413.6	6.5	450	38	414.1	9.7
QX13-C-19.1	0.09	22.5	410	258	0.65	0.0540	2.0	0.4740	2.6	0.0637	1.6	398.1	6.1	372	46	383.3	9.6
QX13-C-21.1	0.00	41.3	750	560	0.77	0.0547	1.1	0.4840	2.8	0.0642	2.6	400.9	9.9	398	25	373.0	10.0
QX13-C-25.1	0.13	42.5	759	626	0.85	0.0542	1.6	0.4870	2.2	0.0651	1.5	406.6	6.0	380	35	386.5	7.1
QX13-C-27.1	--	25.7	453	284	0.65	0.0545	1.4	0.4970	2.1	0.0662	1.6	413.0	6.2	391	32	398.3	9.5
QX13-C-28.1	0.02	41.0	739	639	0.89	0.0539	1.6	0.4800	2.2	0.0647	1.5	403.9	6.0	365	37	383.9	7.1
QX13-C30.1	0.16	16.6	300	209	0.72	0.0544	2.1	0.4840	2.7	0.0646	1.7	403.3	6.8	387	48	386.2	9.1
QX13-C-31.1	0.00	74.8	1369	1448	1.09	0.0541	0.9	0.4741	1.7	0.0636	1.5	397.6	5.7	373	20	374.9	6.4
QX13-C-35.1	--	18.8	334	170	0.53	0.0553	1.6	0.5000	2.2	0.0656	1.6	409.8	6.2	422	35	404.1	8.8
QX13-C-36.1	0.00	17.8	310	157	0.52	0.0557	1.7	0.5110	2.3	0.0666	1.6	415.6	6.4	441	37	413.5	9.3
QX13-C-37.1	0.00	16.5	291	152	0.54	0.0550	1.7	0.4990	2.4	0.0658	1.6	411.0	6.3	412	39	382.1	9.0
QX13-C-40.1	0.08	35.9	654	383	0.61	0.0541	1.5	0.4770	2.1	0.0639	1.5	399.2	5.9	377	33	388.4	7.6
QX13-C-43.1	0.00	20.8	368	172	0.48	0.0546	1.6	0.4950	2.2	0.0657	1.6	410.4	6.2	395	35	399.8	8.7
QX13-C-45.1	--	47.2	858	773	0.93	0.0550	1.1	0.4863	1.9	0.0641	1.5	400.3	5.8	414	25	388.2	6.6
QX13-C-46.1	0.07	34.7	624	553	0.91	0.0546	2.0	0.4870	2.6	0.0647	1.6	404.2	6.4	397	45	388.1	7.9
QX13-C-47.1	--	31.7	568	432	0.79	0.0540	1.3	0.4837	2	0.0650	1.5	406.1	6.0	369	29	380.8	7.0
QX13-C-48.1	--	16.4	286	143	0.52	0.0559	1.8	0.5130	2.4	0.0666	1.6	415.4	6.4	446	41	406.1	9.6
QX13-C-50.1	0.00	41.2	734	708	1.00	0.0538	1.1	0.4842	1.9	0.0653	1.5	408.0	6.0	361	25	388.7	6.6

"--" indicates that ^{204}Pb content of the measured zircon is lower than background
[1] Indicates the common Pb content
[2] Indicates the radiogenic Pb content

Variable weights of powdered samples (3–5 g for the massive, blotchy, and net-textures ores, 5–8 g for the disseminated ores, and 10 g for the sulfide-barren rocks) were used for analysis. The total procedural blanks vary from 0.009 ng (Ir) to 0.033 ng (Pd), and the detection limits range from 0.004 ppb (Ir) to 0.012 ppb (Pd; Qi et al., 2011). The measured results of the CCRMP reference materials TDB-1 and WGB-1 (Table 2) agree well with recommended values reported by Qi et al. (2004, 2011). Precision and accuracy of the measurements are better than 10% when concentrations of PGE are higher than 0.2 ppb and better than 20% when concentrations of PGE are lower than 0.2 ppb. The precision and accuracy are proved by the results of the standards WGB-1 and UMT-1, as well as the data of four duplicate samples listed in Table 2. The relative standard deviations (RSD) of the duplicate samples vary from 1.2 to 9.8% (Table 2).

Results

Zircon U-Pb ages

The zircons in the samples exhibit mostly long or short prismatic morphologies and magmatic zonation (Fig. 8a). Twenty-one zircon grains from QX13-A, 18 grains from QX13-B, and 22 grains from QX13-C were measured by SHRIMP. Most of the zircons contain 320 to 1,800 ppm U and 180 to 1,450 ppm Th and have Th/U ratios of 0.44 to 1.06, with the exception of a few spots and radiogenic ^{206}Pb contents of the zircon grains that range from 18 to 102 ppm (Table 1). The gabbronorite to the north of the ultramafic portion (XQ13-A, Fig. 3b) shows a ^{206}U/^{238}Pb age of 405.5 ± 2.7 Ma (MSWD = 0.98; Fig. 8b). The zircons of the websterite samples of QX13-B and QX13-C yield the mean ^{206}U/^{238}Pb ages of 408.1 ± 2.9 Ma (MSWD = 0.54) and 406.1 ± 2.7 Ma (MSWD = 0.89), respectively, identical within error to the age of the gabbronorite to the north of the ultramafic portion (Fig. 8c, d). These zircon U-Pb ages are coeval with the widespread granitoid plutons in the Middle Kunlun zone (Fig. 2c). Their coincident ages indicate that both the gabbroic and ultramafic portions of the Xiarihamu intrusion formed in the Early Devonian. The gabbronorite xenoliths within the ultramafic portion indicate that the gabbronorite was emplaced slightly earlier. In contrast, LA-ICP-MS data of the zircons separated from the small gabbronorite body occurring to the south of the ultramafic portion (Figs. 3b, 4) yielded an older U-Pb age of 431.3 ± 2.1 Ma (Li et al., 2015). This suggests that the main gabbronorite to the north was emplaced synchronous with the ultramafic portion at 405.5 ± 2.7 Ma, whereas the smaller gabbronorite body to the south were emplaced around 20 m.y. earlier.

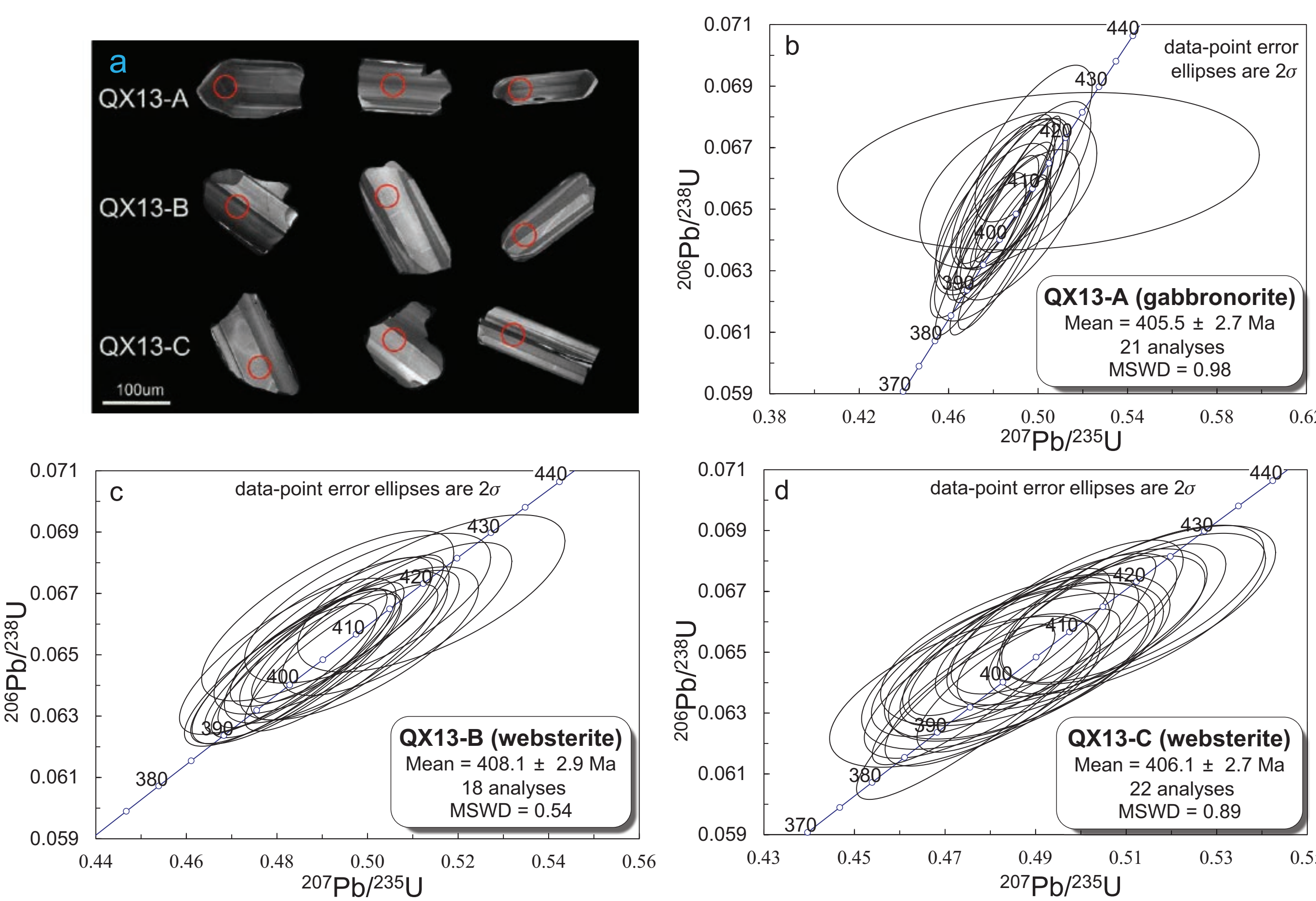

Fig. 8. Cathodoluminescence (CL) images of representative zircons from the Xiarihamu gabbronorite (QX13-A) and websterites (QX13-B and QX13-C) (a); Concordia plots of ID-TIMS zircon U-Pb dating results for the gabbronorite (QX13-A) (b); and the websterites (QX13-B and QX13-C) (c) and (d).

Table 2. Contents of Siderophile and Chalcophile Elements of the Sulfide-Barren Rocks and Sulfide Ores at Xiarihamu

ZK1509												
Sample	QX13-15-68	QX13-15-3	QX13-15-2	QX13-15-64	QX13-15-62	QX13-15-58	QX13-15-55	QX13-15-53	QX13-15-51	QX13-15-49	QX13-15-47	QX13-15-43
Depth (m)	285	483	485	295	304	321	330	336	344	351	358	367
Rock Ore	Gabbronorite	Gabbronorite		Olivine orthopyroxenite		Olivine orthopyroxenite Disseminated ore						
S (wt %)						3.10	0.50	0.90	2.70	3.60	4.50	3.50
Ni (ppm)	184	363	744	1340	1,032	14,100	2,800	4,100	12,800	13,700	18,500	14,000
Cu (ppm)		236	257		117	1,100	400	700	1,800	900	1,800	1,100
Co (ppm)	81.1	106	127	106	105	310	140	170	290	300	380	350
Ir (ppb)		0.01	0.03		0.01			0.08	0.20	0.25	0.47	0.26
Ru (ppb)		0.02	0.05		0.02			0.15	0.30	0.29	0.62	0.33
Rh (ppb)		0.02	0.05		0.02			0.08	0.22	0.27	0.36	0.22
Pt (ppb)		0.37	0.11		0.17			2.73	0.93	0.82	0.91	1.03
Pd (ppb)		1.02	1.21		0.52			1.36	2.99	4.88	7.62	4.62
Ni/Cu[1]		1.5	2.9		8.8	8.8	5.4	5.0	6.8	14.5	9.9	12.2

ZK1509												
Sample	QX13-15-38	QX13-15-35	QX13-15-33	QX13-15-29	QX13-15-24	QX13-15-24[2]	QX13-15-19	QX13-15-17	QX13-15-14	QX13-15-8	QX13-15-5	QX13-15-4
Depth (m)	389	401	410	423	436		445	449	459	472	479	481
Rock Ore	Olivine orthopyroxenite Disseminated ore											Gabbronorite Disseminated ore
S (wt %)	3.10	3.30	3.80	5.30	3.00		3.80	4.50	3.80	1.80	2.60	3.30
Ni (ppm)	12,800	15,400	15,100	29,700	13,600		14,600	18,400	17,200	5,200	9,900	11,800
Cu (ppm)	2,200	3,600	14,400	2,800	1,400		1,600	1,500	2,500	400	1,200	1,500
Co (ppm)	250	260	280	510	310		360	420	330	160	260	260
Ir (ppb)	0.03	0.05	0.07	0.29	0.24	0.24				0.12	0.29	
Ru (ppb)	0.02	0.05	0.04	0.27	0.31	0.36				0.20	0.36	
Rh (ppb)	0.09	0.12	0.13	0.30	0.23	0.25				0.06	0.26	
Pt (ppb)	7.27	1.64	2.19	1.01	0.54	0.61				0.25	45.8	
Pd (ppb)	7.18	5.93	12.43	12.20	5.26	5.17				1.44	3.42	
Ni/Cu	5.8	4.3	1.0	10.4	9.3		8.7	11.9	6.6	11.4	7.7	7.5

Table 2. *(Cont.)*

ZK703													
Sample	QX13-7-15	QX13-7-42	QX13-7-68	QX13-7-17	QX13-7-34	QX13-7-52	QX13-7-84	QX13-7-86	QX13-7-88	QX13-7-91	QX13-7-93	QX13-7-94	QX13-7-95
Depth (m)	223	186	118	217	197	165	72	65	55	40	26	14	9
Rock	Websterite										Gabbronorite		
Ore													
S (wt %)													
Ni (ppm)	388	333	542	289	299	351	344	262	344	334	390	167	75
Cu (ppm)	44.7	48.3	147.5	59.1	52.4	47.4	154	64.6	89.2	62.3	81.6	34.0	15.9
Co (ppm)	82	102	102	88	85	116	97	107	111	108	108	93	102
Ir (ppb)	0.00	0.00	0.01	0.01		0.01	0.01	0.01	0.03	0.01	0.01	0.01	
Ru (ppb)	0.02	0.01	0.02	0.02	0.01	0.01	0.02	0.04	0.04	0.02	0.02	0.01	0.01
Rh (ppb)	0.01	0.00	0.01	0.01	0.00	0.01	0.02	0.02	0.02	0.01	0.02	0.01	0.01
Pt (ppb)	0.08	0.02	0.06	0.21	0.12	0.10	0.27	0.28	0.29	0.24	0.23	0.10	0.08
Pd (ppb)	0.11	0.06	0.53	0.36	0.13	0.07	0.53	0.37	0.56	0.39	0.51	0.18	0.13
Ni/Cu	8.7	6.9	3.7	4.9	5.7	7.4	2.2	4.1	3.9	5.4	4.8	4.9	4.8

ZK703												
Sample	QX13-7-13	QX13-7-23	QX13-7-23[2]	QX13-7-30	QX13-7-37	QX13-7-44	QX13-7-50	QX13-7-60	QX13-7-66	QX13-7-66[2]	QX13-7-70	QX13-7-72
Depth (m)	228	207		199	191	184	170	144	123		114	109
Rock	Orthopyroxenite											
Ore	Massive ore											
S (wt %)	29.0	27.6		21.8	30.5	22.7	25.5	22.4	24.8		28.7	22.8
Ni (ppm)	41,700	35,800		28,100	34,300	28,200	32,800	19,200	38,100		41,500	22,200
Cu (ppm)	1,500	1,400		2,100	500	1,600	1,700	1,400	2,700		2,400	900
Co (ppm)	2,210	1,980		1,620	2,000	1,720	2,030	1,100	2,140		2,340	1,160
Ir (ppb)	1.44	0.97	1.05	1.17	1.56		1.19		1.32	1.21		1.18
Ru (ppb)	2.18	2.12	1.96	1.64	2.38		1.97		1.81	1.77		1.71
Rh (ppb)	0.89	2.56	2.49	1.09	1.76		1.00		1.25	1.31		1.15
Pt (ppb)	0.29	9.40	10.8	0.29	0.11		1.93		0.12	0.11		0.32
Pd (ppb)	40.0	24.5	24.9	37.5	36.1		34.9		27.4	26.8		16.5
Ni/Cu	27.8	25.6		13.4	68.6	17.6	19.3	13.7	14.1		17.3	24.7

ZK703										
Sample	QX13-7-18	QX13-7-25	QX13-7-12	QX13-7-32	QX13-7-35	QX13-7-59	QX13-7-39	QX13-7-47	QX13-7-11	QX13-7-22
Depth (m)	217	205	230	199	195	145	188	179	236	207
Rock	Orthopyroxenite									
Ore	Blotchy or net-textured ore									
S (wt %)	20.0	19.7	10.9	16.2	19.8	10.6	9.90	3.60	3.20	6.40
Ni (ppm)	25,700	29,000	15,100	21,800	23,200	14,100	11,400	4,500	7,000	8,700
Cu (ppm)	1,100	4,200	2,000	2,500	400	3,800	37,500	3,400	1,200	1,800
Co (ppm)	1,230	1,850	730	1,290	1,350	850	740	290	310	540
Ir (ppb)		0.92	0.75	0.82	0.96		0.44		0.03	0.31
Ru (ppb)		1.46	1.29	1.14	1.46		0.71		0.07	0.53
Rh (ppb)		1.18	1.55	0.76	1.68		0.60		0.12	0.36
Pt (ppb)		0.21	11.51	0.30	0.09		0.14		0.51	0.26
Pd (ppb)		30.14	13.28	26.22	6.93		11.49		7.34	10.29
Ni/Cu	23.4	6.9	7.6	8.7	58.0	3.7	0.3	1.3	5.8	4.8

Table 2. (*Cont.*)

ZK703														
Sample	QX13-7-7	QX13-7-81	QX13-7-54	QX13-7-4	QX13-7-16	QX13-7-5	QX13-7-20	QX13-7-51	QX13-7-63	QX13-7-71	QX13-7-73	QX13-7-75	QX13-7-77	QX13-7-79
Depth (m)	247	85	160	254	221	252	212	167	130	112	106	100	94	90
Rock	Orthopyroxenite													
Ore	Disseminated ore		Sparsely disseminated ore											
S (wt %)	3.50	3.10	3.40	2.00	1.80	1.60	1.90	1.10	1.70	1.20	1.30	1.90	1.20	1.80
Ni (ppm)	6,200	4,500	3,900	3,800	2,900	2,900	2,700	1,400	2,300	2,000	2,000	3,600	2,400	3,200
Cu (ppm)	700	700	700	800	500	900	300	1,200	300	400	300	500	600	700
Co (ppm)	340	220	300	230	230	190	190	160	180	180	170	230	190	210
Ir (ppb)		0.09	0.16	0.08	0.13			0.03				0.14	0.04	0.05
Ru (ppb)		0.17	0.24	0.18	0.20			0.26				0.25	0.12	0.12
Rh (ppb)		0.13	0.13	0.13	0.08			3.24				0.11	0.10	0.08
Pt (ppb)		0.50	0.01	0.07	0.89			0.16				0.76	0.44	1.22
Pd (ppb)		3.10	4.24	2.37	2.56			0.66				3.35	1.51	1.96
Ni/Cu	8.9	6.4	5.6	4.8	5.8	3.2	9.0	1.2	7.7	5.0	6.7	7.2	4.0	4.6

ZK203													
Sample	QX13-2-27	QX13-2-23	QX13-2-19	QX13-2-17	QX13-2-1	QX13-2-13	QX13-2-9	QX13-2-7	Average				
Depth (m)	41	76	108	127	211	147	171	183	blank[3]	WGB-1	WGB-1	UMT-1	UMT-1
Rock	Orthopyroxenite					Orthopyroxenite				Expected[4]	This study (mean ± S)[5]	Expected[4]	This study (mean ± S)[5]
Ore						Disseminated ore	Blotchy textured ore		N = 4		N = 4		N = 4
S (wt %)						2.10	8.80	5.90					
Ni (ppm)	293	577	210	258	258	2,300	8,300	6,700					
Cu (ppm)	36.0	163	12.9	26.9	25.9	500	1500	900					
Co (ppm)	93.3	91.2	67.0	111	95.2	180	630	540					
Ir (ppb)	0.00	0.02	0.00	0.01	0.01	0.11	0.44	0.28	0.002	0.27 ± 0.03	0.20 ± 0.02	8.23 ± 0.35	7.46 ± 0.30
Ru (ppb)	0.02	0.03	0.01	0.02	0.02	0.18	0.58	0.39	0.005	0.15 ± 0.02	0.13 ± 0.01	9.83 ± 0.82	8.74 ± 0.13
Rh (ppb)	0.00	0.02	0.01	0.01	0.02	0.11	0.44	0.31	0.001	0.18 ± 0.02	0.20 ± 0.01	8.89 ± 0.7	10.3 ± 0.54
Pt (ppb)	0.05	0.13	0.11	0.11	0.13	0.11	0.05	0.21	0.017	4.95 ± 0.52	4.46 ± 0.49	135 ± 6	136 ± 12
Pd pp)	0.14	0.55	0.09	0.16	0.90	1.28	11.8	12.1	0.008	11.8 ± 0.8	12.2 ± 0.99	108 ± 5	109 ± 9
Ni/Cu	8.1	3.5	16.2	9.6	10.0	4.6	5.5	7.4					

[1] Before Ni/Cu ratio calculation, Ni in olivine had been deducted from the whole-rock Ni contents; average modal percentage of olivine in the disseminated ore (olivine orthopyroxenites) is ~25% and average Ni content of the olivine is ~2,500 (ppm), according to our observation and electronic microprobe data of ours and Li et al. (2015)
[2] Duplicate samples
[3] Blank value is expressed as concentrations in 10-g sample
[4] Expected data of the WGB-1 and UMT-1 are after Qi et al., 2011
[5] S = standard deviation

Siderophile and chalcophile element compositions

The sulfide-barren orthopyroxenites and gabbronorite contain less than 1,000 ppm Ni and 200 ppm Cu (Table 2). Positive correlation between Ni and Cu suggests that the metals are mostly hosted in minor sulfides (not shown). Electron microprobe data indicate that the olivine grains of the olivine orthopyroxenite contain 1,690 to 2,300 ppm Ni with 85 to 88 mol % forsterite (Fo; our unpub. data). This is consistent with the recently published data of Li et al. (2015) of the olivine in most of the rocks containing trace sulfides at Xiarihamu, although the olivines hosted in disseminated sulfide ores commonly are higher in Ni. This is consistent with the observation that the sulfide-barren olivine orthopyroxenite contains more than 1,000 ppm whole-rock Ni (Table 2).

The sulfide ores of M1 at Xiarihamu have average grades of 0.65 wt % Ni, 0.14 wt % Cu, 0.013 wt % Co (No. 5 GMSI), and a large variation in Ni (0.1–4.2 wt %) and Co (up to 0.23 wt %). The sulfide ores are characterized by extremely low Cu (<0.5 wt %) and PGE (<40 ppb Pd, <45 ppb Pt, and < 2 ppb Ir) and unusually high Ni/Cu ratios (ranging from 4–69, with only a few exceptions). The disseminated sulfides hosted in the olivine orthopyroxenites (from ZK1509) have Ni/Cu ratios varying from 5.0 to 14.5 (with exception of one sample) and an average of 9.0. The Ni/Cu ratio calculation takes account of an estimated background silicate Ni content, based on median Ni content of olivine (~2,500 ppm) according to electronic microprobe data of ours and Li et al. (2015) and average modal percentage of olivine in the disseminated ore (olivine orthopyroxenites; ~25%, see above). The disseminated sulfides hosted in the orthopyroxenites (from ZK703 and ZK203) have slightly low Ni/Cu ratios, ranging from 3.0 to 9.0 except one sample. The massive sulfides (from ZK703) have the highest Ni/Cu ratios (13.4–27.8, except one sample; Table 2). Such Ni/Cu ratios are evidently higher than those of the Ni-Cu sulfide deposits related to basaltic magma, such as Jinchuan, Noril'sk, and Voisey's Bay (Naldrett, 2004), which typically have Ni/Cu ratios of less than 2. The positive correlations between sulfur and the metals (Ni, Co, Ir, Ru, Rh, and Pd) for the sulfide ores indicate that the siderophile and chalcophile elements are dominantly contained in the sulfides, although Ni and Co are also compatible in olivine and pyroxene (Fig. 9). Palladium is perfectly positively correlated with S in the sulfide ores, but Pt does not show any correlation with S (Fig. 9g, h).

The massive, blotchy, and net-textured ores are high in Ni, Co, and PGE in whole rock (except for Pt) because of their high sulfide contents, whereas they are low in Cu and high in Ni/Cu ratios (Table 2), ranging from 5 to 68, relative to the disseminated ores with Ni/Cu ratios of 3 to 15 (three outlying samples excluded). These Ni/Cu ranges in sulfide are calculated allowing for Ni contained by olivine in the olivine orthopyroxenites using olivine percentages and average Ni content of the olivine as described above. The massive, blotchy and net-textured ores are characterized by small variation in PGE in 100% sulfides, except for Pt, relative to the disseminated ores (Fig. 10). The massive, blotchy and net-textured ores have relatively small variation in Pd/Ir ratio (= 14–32), whereas Pd/Ir ratios of the disseminated sulfide ores range from 10 to 260. As shown in Figure 10, the disseminated ores (containing less than 15% sulfides, ZK1509) hosted by the olivine orthopyroxenites have higher Ni tenors (13.0–19.5 wt %) than the massive net-textured and disseminated ores hosted in the orthopyroxenites (Ni_{100} = 3.6–8.2 wt % in ZK703 and ZK203, the subscript "$_{100}$" means metal content in 100% basis sulfide, calculated according to Barnes et al. (2011) and Song et al. (2008)), whereas Cu_{100} of all types of the ores are clearly lower than Ni_{100} and less than 4.2 wt % with the exception of only one sample (Figs. 11, 12b, Table 2).

It is notable that the disseminated sulfides from the western part of M1 (drill hole ZK1509) have obviously higher Ni contents than those from the middle and eastern parts (drill holes ZK703 and ZK203) and are similar in the other metals at comparable S contents (Fig. 9). Some of the disseminated ores in both western and eastern parts of M1 display depletions of Ir and Ru. Although Ir_{100} is well positively correlated with Ru_{100} and Rh_{100}, no correlation appears between Pt_{100} and Pd_{100} (Fig. 10a-d). Especially, Pt_{100} values of the massive, blotchy and net-textured ores are much lower than those of the disseminated ores (Fig. 10c, e). The extremely large variations of the disseminated ores in the ratios of Pd/Ir (= 10–260) and Pd/Pt (= 0.5–60) demonstrate extensive Pd-Ir and Pd-Pt decoupling, particularly for the massive and net-textured ores (Fig. 10d-e). Figure 10f-h show that the disseminated sulfides from the drill hole ZK1509 are evidently higher in Ni_{100} and Pt_{100} and have larger range of Ir_{100} than those from the drill holes ZK703 and ZK203.

Discussion

Associated magmatism

Extensive Late Silurian and Early Devonian magmatism occurred in the middle East Kunlun zone, producing huge granitoid plutons and relatively small gabbroic intrusions as well as volcanic sequences (Fig. 2). More than 600 zircon U-Pb age data measured recently using LA-ICPMS and SHRIMP-II indicated that the Paleozoic granitoids in the East Kunlun zone were dominantly formed between 450 and 390 Ma and show two peak ages, 433 and 405 Ma, respectively (Fig. 2c; Liu et al., 2012). The granitoids in the west part of the middle East Kunlun zone are slightly older (440–420 Ma, with a peak age of 433 Ma) than those in the east part (420–400 Ma, with a peak age of 405 Ma). Occurrences of early Paleozoic ophiolite fragments and ~430 Ma eclogites along the western parts of the central East Kunlun fault were regarded as the evidence for collision between the middle and southern East Kunlun microcontinents in the Late Silurian (Fig. 2b; Jiang et al., 1992; Yang et al., 1999, 2000; Chen et al., 2002). Therefore, the Late Silurian granitoid plutons (440–420 Ma) in the western portion of the middle East Kunlun zone are attributed to partial melting of crust due to underplating of mafic magmas derived from the upper mantle (Liu et al., 2012). In the east part of the middle East Kunlun zone, it is remarkable that mafic-ultramafic intrusions dated at 410 to 400 Ma occur along with the Early Devonian granitoids and the synchronous Maoniushan Formation rhyolites (408.2 ± 2.4 Ma, Lu et al., 2010; Liu et al., 2012; Fig. 2d). The consistent ages indicate a close genetic relationship between the Early Devonian mafic-ultramafic intrusions and the granitoids. The occurrences of the mafic-ultramafic intrusions clearly demonstrate

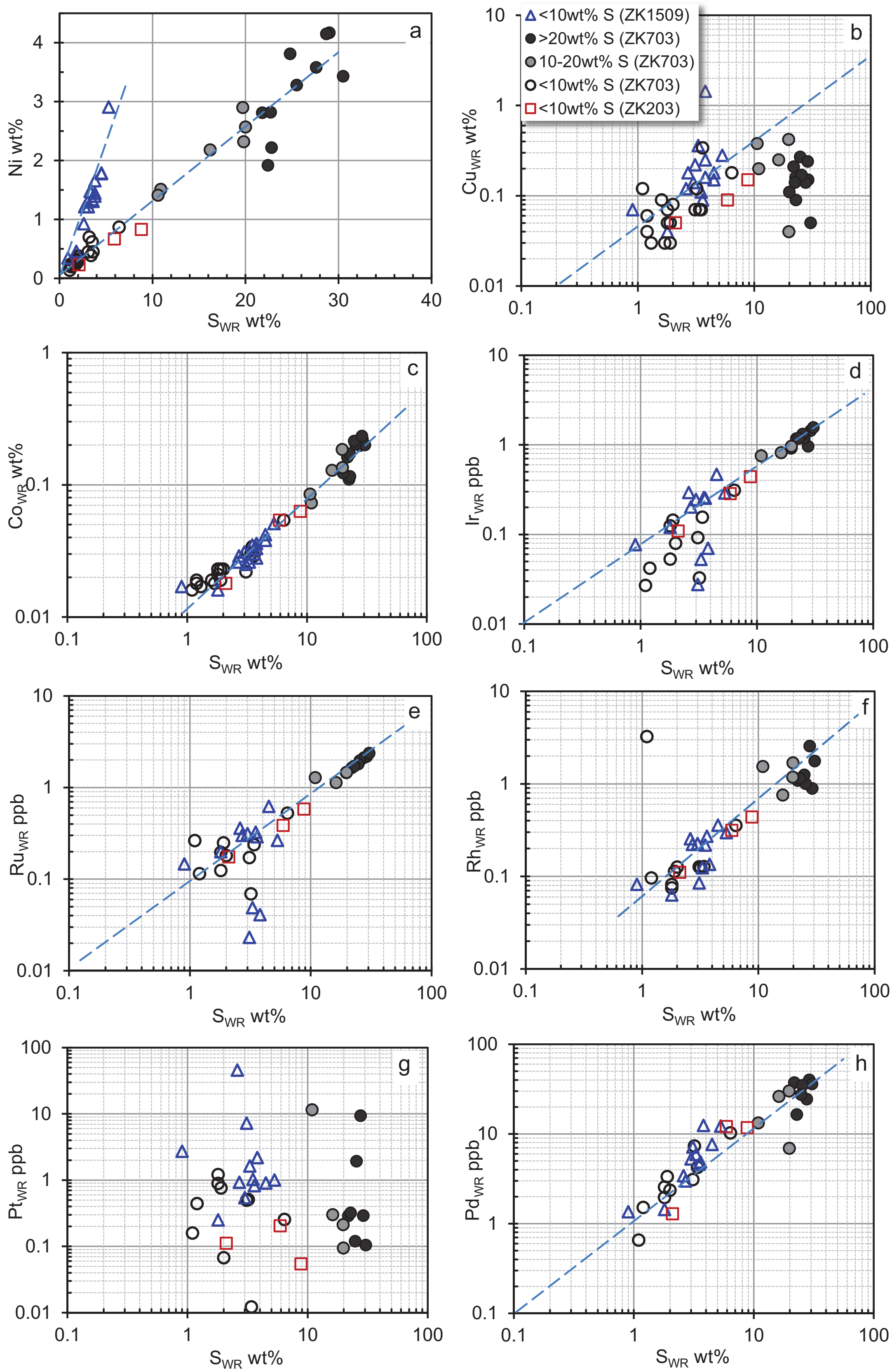

Fig. 9. Whole-rock elemental binary plots of the sulfide ores at Xiarihamu. Locations of the drill holes ZK1509, ZK703, and ZK203 are shown in Figures 3b and 4c, d. The legend caption "<10 wt % S (ZK1509)" indicates samples from bore hole ZK1509 containing less than 10 wt % sulfur.

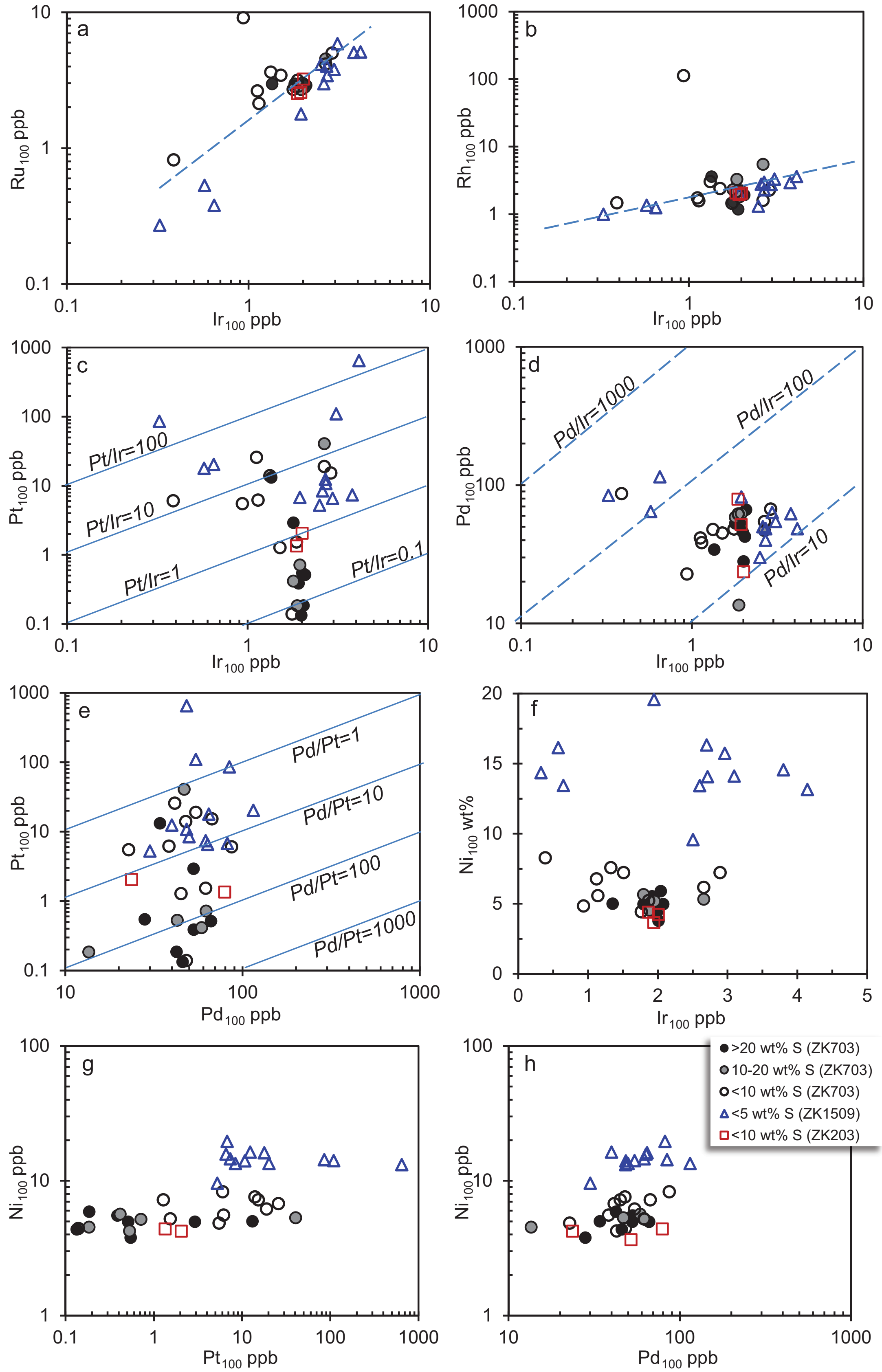

Fig. 10. Binary plots of the siderophile and chalcophile elements of the sulfide ores at Xiarihamu in 100% sulfide basis.

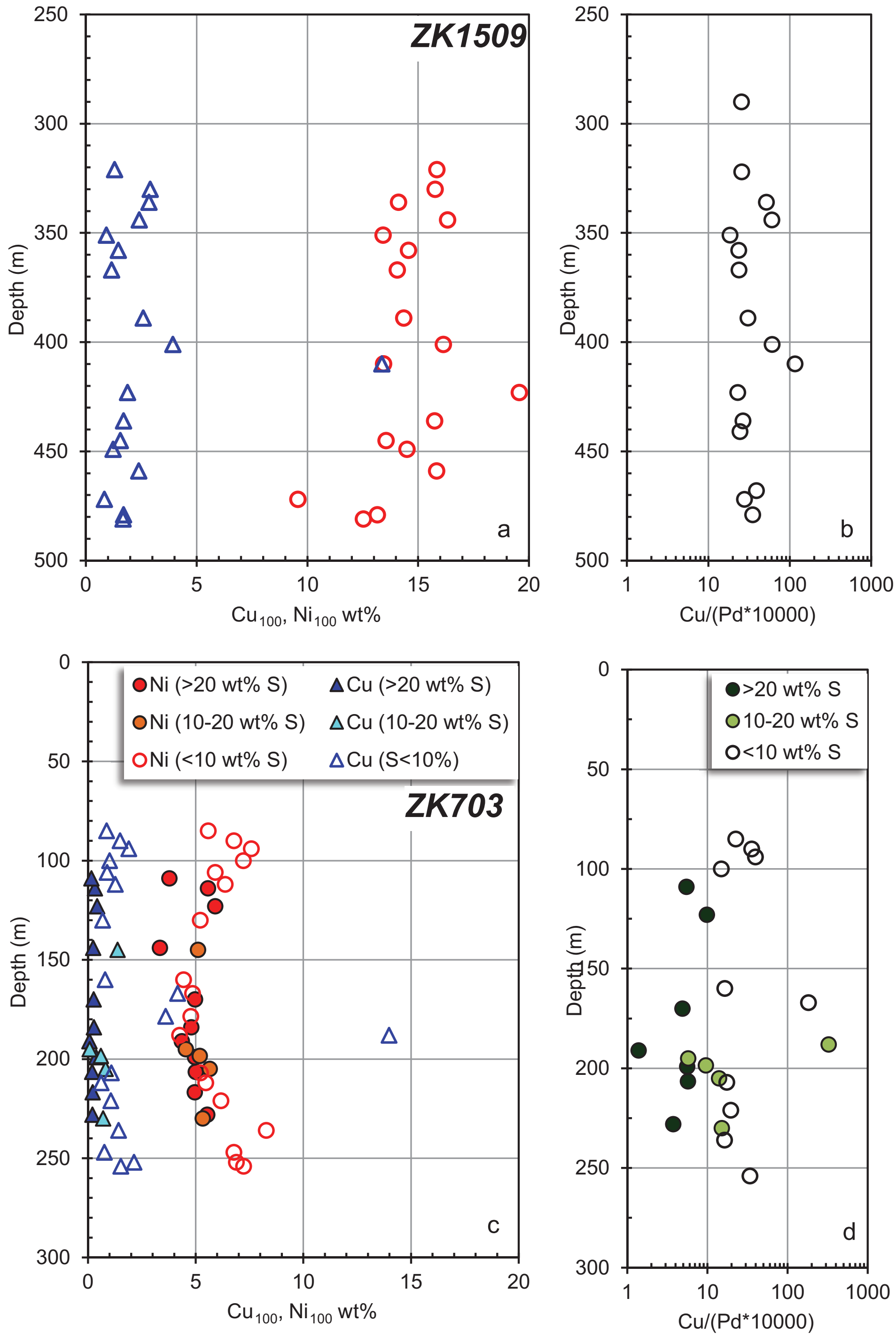

Fig. 11. Stratigraphic variations of Ni and Cu in 100% sulfide basis as well as Cu/Pd ratio of the drill holes of ZK703 and ZK1509.

the emplacement of mantle-derived magmas in the crust. Low ($Na_2O + K_2O$) values (= 6.45–8.40 wt %) and 10^{4*}Ga/Al ratios (=2.1–2.5), low $\varepsilon_{Nd(t)}$ (–3.5 to –5.8) and high $(^{87}Sr/^{86}Sr)_i$ (0.710–0.740), as well as high crystallization temperature of zircons (>800°C) separated from the granitoids indicate that they are I- and S-type granite and were generated by crustal melting (Liu et al., 2012). The enormous Early Devonian granitoid plutons as well as the synchronous mafic-ultramafic intrusions in the east part of the middle East Kunlun zone indicate underplating of a huge amount of mantle-derived basaltic magmas, which resulted in extensive crustal melting. Occurrence of a terrestrial red molasse layer up to 200 m thick at the base of the Early Devonian Maoniushan Formation, which uncomfortably overlies the late Ordovician strata, implies a regional uplifting before the extensive magmatism. Additionally, a few ~391 Ma syenite intrusions with affinities

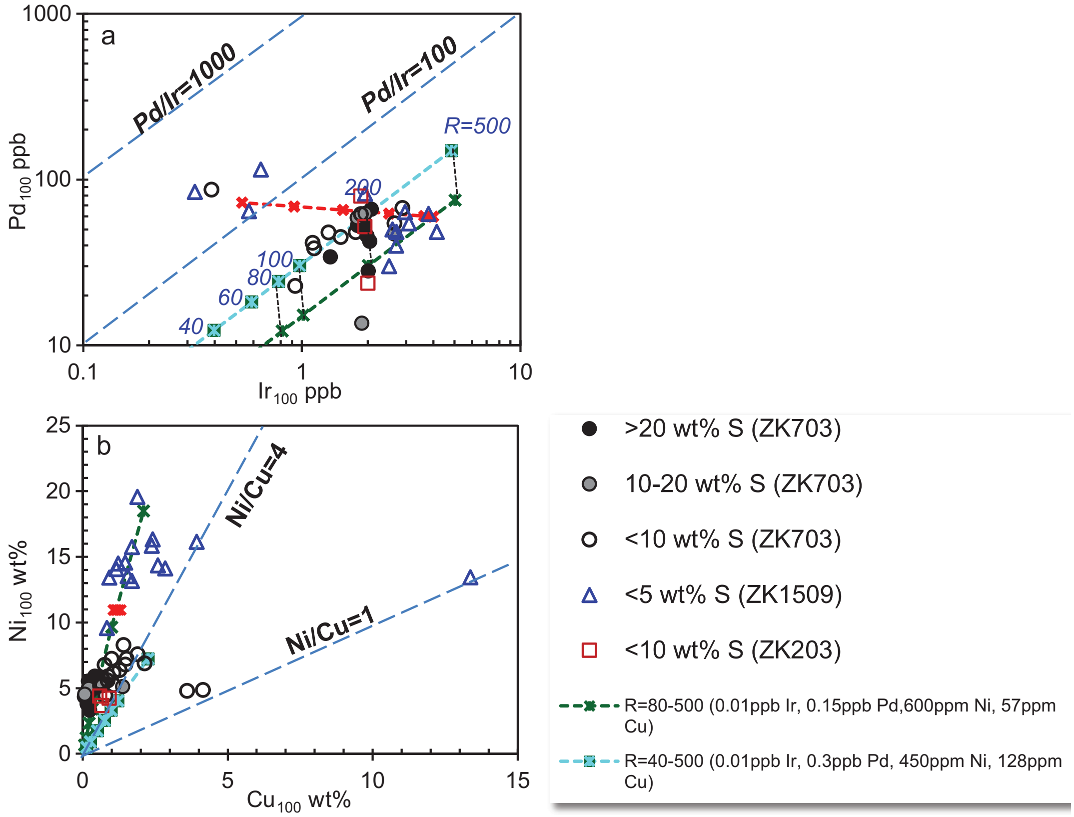

Fig. 12. (a). Plots of Ir_{100} vs. Pd_{100} show larger variation of disseminated sulfides in Ir tenor and Pd/Ir ratio than the massive and net-textured sulfides. (b). Plots of Cu_{100} vs. Ni_{100}, indicating differences between Cu and Ni tenors of the sulfides in the eastern and western parts of the orebody M1. Model calculations indicate the differences of the sulfides in the eastern and western part of M1 in relationship to parental magma composition and R values, consistent with the results shown in Figure 13.

of A_2-type granite, indicated by high 10^4*Ga/Al ratios (>3.0) and characteristic Nb-Y-Ce relationships, have been discovered in the east part of the middle East Kunlun zone (Fig. 2d; Eby, 1992; Liu et al., 2013; Wang et al., 2013).

The ages of the Xiarihamu gabbronorite and websterites (~406 ± 3 Ma, Fig. 8) are consistent with the peak ages of ~405 Ma of the extensive regional magmatism in the east part of the middle East Kunlun zone (Figs. 2, 3). The evidence indicates that the formation of the Xiarihamu deposit is closely related to the extensive partial melting of the upper mantle in Early Devonian. Most of the recent studies proposed that the Late Silurian to Early Devonian magmatism in the East Kunlu zone occurred in a syn- or postcollisional setting. Li et al. (2015) proposed that the Xiarihamu deposit was formed in an arc setting according to their boninitic parental magma estimation (9.8 wt % MgO and 52.4 wt % SiO_2) based on the compositions of the lherzolite and hosting olivine with forsterite percent ranging from 88 to 89.3 mol % and Ca depletion. However, the spinels enclosed in the olivine crystals of the Xiarihamu ultramafic rocks contain low Cr_2O_3 contents (20.4–30.7 wt %) and high Al_2O_3 contents (34.5–44.4 wt %; Li et al., 2015) relative to the spinel in boninite, which is characterized by high Cr_2O_3 (45–60 wt %) and low Al_2O_3 (<15 wt %; Zhou et al., 1998; Pagé and Barnes, 2009; Akmaz et al., 2014). Primary boninitic magma is believed to be generated from remelting of depleted residual peridotite and is commonly S undersaturated and high in PGE relative to tholeiitic magma (Crawford et al., 1989; Zhou et al., 1998). Thus, the sulfides segregated from the boninitic magma should be PGE undepleted in the absence of prior sulfide removal. More work is needed to characterize the parent magmas of the Xiarihamu intrusion.

It is thought likely that the upper mantle had been metasomatized by fluid or melt from the subduction slab. Similarly, the Early Permian mafic-ultramafic intrusions hosting economic Ni-Cu deposits in the Huangshan-Jingerquan belt at the southern margin of the Central Asian orogenic belt were also genetically related to partial melting of metasomatized mantle source (Song et al., 2013).

Sulfide deposition constrained by magma chamber geometry

Figures 3c and 6 show that the pinch-and-swell of the M1 orebody depends strongly on the geometry of the ultramafic portion of the Xiarihamu intrusion. The shapes, sulfide contents, and ore textures of M1 on the two sides of exploration line 9 are clearly different as described above. The two thick ore pods are dominantly comprised of disseminated ores containing less than 20% sulfides. They correspond very well with the thickest parts of the ultramafic sequence to the west of exploration line 9 (Figs. 3c, 6). These features lead us to propose that the two thick ore pods were formed by settling of olivine and sulfides when the slurry carrying olivine crystals and sulfide droplets entered the Xiarihamu intrusion from west side as the channel expanded abruptly. The slurry flow was divided into two flow branches because of the occurrence of the NE-SW-trending discontinuity between exploration lines 17 and 15 (Figs. 3c, 4d). The northern branch flowed to the east and then to the south to form the larger thick ore pod in the east of the discontinuity, whereas the southern branch flowed to the south to form the smaller ore pod situated to the west of the discontinuity (Fig. 6). Settling of sulfide droplets and olivine grains from the silicate magma resulted in the southward decrease of sulfide contents (Fig. 6). Similar processes have also been proposed to explain the concentration of sulfides in the Voisey's Bay Ni-Cu-Co deposit (Evans-Lamswood et al., 2000; Li et al., 2001). Additionally, the orebody becomes very thin and poor in sulfides to the south of the NW-SE-trending discontinuity between lines 19 and 17, which prevented the slurry flowing farther south. The $\delta^{34}S$ values of the sulfides ores ranging from 3.5 to 6.8 indicate that sulfide liquid immiscibility was triggered by crustal sulfur addition before the slurry intruded into the Xiarihamu intrusion (Li et al., 2015), whereas local S derivation is unlikely because the wall rocks are poor in sulfide. This implies that the sulfides were carried from deeper levels. The rounded edges of the olivine crystals in the olivine orthopyroxenite containing sulfides again imply the olivine was carried from depth and resorbed due to the reaction between olivine and magma to form the orthopyroxene.

Merging toward the north of the ore sublayers east of exploration line 7 indicates that the slurries containing sulfides were emplaced from the west (Fig. 6). However, absence of olivine in the orthopyroxenite hosting the sulfides, the occurrences of the massive, net-textured and blotchy textured ores and their low Ni tenors suggest that the ore sublayers in the middle and eastern parts of the ultramafic portion were formed by distinct pulses of magma slurries containing more sulfide droplets. This probably implies the middle and eastern parts of the ultramafic portion opened slightly earlier than the western part. The decrease in thickness and sulfide contents of the ore sublayers toward the southeast can be attributed to extensive decrease of capacity to hold sulfide droplets suspension in the slurry due to slowing of the flow when the slurry entered the capacious space to the east of line 17 (Fig. 4b). The sulfide droplets could have coalesced and percolated through the cumulus pile to form the blotchy textured ores, or to be concentrated in embayments at the bottom of the ore layer to form massive ores (Chung and Mungall, 2009).

Implications of compositional diversity of the sulfides

Siderophile and chalcophile element concentrations of magmatic sulfides depend on silicate/sulfide mass ratio (R factor) during sulfide segregation as well as concentrations of these elements in the parental magma, with additional variance due to fractionation of the sulfide liquid in some circumstances. Concentrations of these elements in the parental magma are related to the nature of the mantle source, degree of partial melting, and magma fractionation. Additionally, these elements in the parental magma may be enriched due to dissolution of previously formed sulfide liquids or depleted by prior sulfide removal (Kerr and Leitch, 2005; Song et al., 2008, 2009b). The segregated sulfides may be upgraded in the siderophile and chalcophile elements via reaction with new S-undersaturated magma (Kerr and Leitch, 2005; Song et al., 2008) and enriched in Ni due to reaction with olivine crystals (Barnes et al., 2013).

Copper, Pd, and Pt are considered able to be transported in solution during hydrothermal alteration as bisulfide or chloride complexes such as $CuCl_2^-$, $AuCl_2^-$, $PtCl^{3-}$, and $PdCl_4^{2-}$ under reduced and moderate oxidation or extremely acidic/oxidized condition in temperature between 500° and 300°C (Gammons et al., 1992; Xiong and Wood, 2000; Wood, 2002; Liu and McPhail, 2005; Barnes and Liu, 2012). Hydrothermal alteration can account for the distinctive compositions of disseminated sulfide ore samples characterized by very high Pd/Ir ratios (>100, up to 260) at Xiarihamu (Fig. 10c, d). The clearly lower Ir and Ru contents of these samples than the other samples under comparable sulfur contents (Fig. 9d, e) suggest that precipitation of Pd took place in these disseminated sulfide ores during hydrothermal alteration as a result of breakdown of bisulfide or chloride complexes. This process resulted in increase of Pd rather than Ir and Ru and led to very high Pd/Ir ratios (Fig. 10c, d). However, decoupling between Pt and Pd shown by large variation of Pd/Pt ratios cannot be attributed to hydrothermal alteration only, although Pd is typically an order of magnitude more soluble than Pt as bisulfide or chloride in hydrothermal fluid. The compositional features and variation of siderophile and chalcophile elements of the sulfide ores at Xairihamu as described above can be attributed to the following processes.

PGE-depleted basaltic magma derived from pyroxenite mantle: The most striking characteristics of the Xiarihamu sulfide ores are their unusually high Ni/Cu ratios and extremely low PGE tenors (Ir_{100} <4.1 ppb, Pt_{100} <110 ppb, Pd_{100} <115 ppb, with exception of one sample; Table 2, Fig. 10). Although the disseminated ores containing less than 10 wt % sulfur have very high Ni/Cu ratios (= 3–14.5, except for two samples), similar to those of komatiite-related deposits (= 4–58) and higher than those of flood basalt-related deposits (<2; Naldrett, 2004), they are depleted in PGE (Figs. 9, 10). The remarkable high Ni/Cu ratios of komatiite-related sulfides are believed to be a necessary sequence of initial Ni and PGE enrichment of komatiitic magma generated by a very high degree partial melting of the mantle peridotite (Barnes, Sarah-J. et al., 1982; Barnes, Stephen J. et al, 1995; Lesher and Campbell, 1993; Arndt et al., 2005, 2008). However, Cu and PGE tenors of the Xiarihamu sulfides are not only much lower than those of the Aguablanca (Spain) and Santa Rita (Brazil) Ni-Cu-(PGE) deposits (Piňa et al., 2008; Barnes et al., 2011), but also evidently lower than the sulfides segregated from PGE-depleted basaltic magmas in the Ni-Cu deposits at the southern margins of the Central Asian orogenic belt (Table 3, Song and Li, 2009; Deng et al., 2014; Mao et al., 2014). This demonstrates that the high Ni/Cu ratios of the Xiarihamu sulfides cannot be attributed to high-degree partial melting of the upper mantle. Thus, the high Ni/Cu ratios of the Xiarihamu sulfides are attributed to the parental magma being strongly enriched in Ni relative to Cu, because the partition coefficient of Cu between sulfide liquid and silicate melt ($D_{Cu}^{Sul/Sil}$ = ~1,400) is slightly higher than $D_{Ni}^{Sul/Sil}$ (= ~500–900; Peach et al., 1990). Although olivine could release a small amount of Ni during reaction

Table 3. Siderophile and Chalcophile Element Compositions of the Ni-Cu Sulfide Deposits in Orogenic Belts

Metal content in 100% sulfide	For both disseminated and massive sulfide ores							For disseminated ores (S <10 wt %)				
	Ni (wt %)	Cu (wt %)	Ir (ppb)	Ru (ppb)	Rh (ppb)	Pt (ppb)	Pd (ppb)	Cu/(Pd*10^4)	Ni/Cu	Pd/Ir	Pd/Pt	Ni/Co
Xirihamu	3.7–19.9	0.1–4.2	0.32–4.14	0.27–9.13	1–2.66	0.14–85.4	13.6–115	1.4–287	4.0–14.5	11.7–260	0.1–345	8.8–62.8
Huangshandong, CAOB	1.2–11.2	0.1–10.3	0.3–24.3	0.1–61.5	0.5–21.8	1.1–408	6.5–350	2.4–448	0.3–6.3	1.6–178.1	0.3–4.6	
Huangshan, CAOB	1.8–10.4	0.4–16	0.7–8.7	1–40.7	0.7–16.7	1.1–170	7.9–159	15.5–769	0.2–8.1	4.7–29.5	0.4–2.2	
Kalatongke, CAOB	3.1–26.9	1.0–9.7	0.79–63.1	1.65–72.3	1.47–68	127–1,940	30.2–2,071	4.5–85.6	0.1–1.9	9.3–108	0.3–8.4	10.5–29.5
Heishan, CAOB	5.4–7.3	0.1–10.3	48.9–164	85.7–300	59.8–309	584–4,105	725–5,841	1.7–7.5	1.1–1.7	7.5–35.6	0.7–1.7	16–46
Aguablance, Spain	1.7–10.9	0.3–23.9	19.4–318	29.6–196	23.7–328	43.2–5,552	604–5,382	0.2–39.2	0.1–1.3	2.0–234	0.2–8.2	16.9–41.8
Santa Rita, Brazil	10.0–22.0	0.5–10	<3,000	<4,500	<2000	200–10,000	300–8,000	0.01–50	1.0–16	~3–6	0.1–5	

Notes: Data of the Xiarihamu deposit are from this study, data of the deposits in the CAOB (Central Asian orogenic belt) are after Song and Li (2009), Zhang et al. (2011), Gao et al. (2013), Deng et al. (2014), and Xie et al. (2014), data of the Aguablance deposit are after Piña et al. (2008), data of the Santa Rita deposit are from Barnes et al. (2011)

with sulfide liquid (Barnes et al., 2013), the disseminated sulfides hosted in both the olivine orthopyroxenites and the orthopyroxenites have very high Ni/Cu ratios, 5 to 14.5 and 3 to 9, respectively, as described above (Table 2), too high to be explained by sulfide-olivine reaction.

Although partial dissolution of previously segregated sulfide liquids can also cause Ni enrichment in the silicate magmas, this mechanism should cause enrichments of Cu and PGE as well (Kerr and Leitch, 2005). However, the Xiarihamu sulfides are rich in Ni but low in PGE and Cu, causing us to reject this hypothesis. An alternative explanation is derivation from unusually Ni rich magma. The olivines in the rocks containing trace sulfides and disseminated sulfide ores at Xiarihamu have a large range of forsterite percentages (Fo = 83.9–89.3, mostly varying from 86 to 88) and Ni contents (= 1,290–3,760 ppm, mostly varying from 2,000–3,000 ppm; Li et al., 2015). The Ni contents of the Xiarihamu olivines are higher than the olivine with comparable Fo values crystallized from the basaltic magma generated by melting of fertile peridotite (Sobolev et al., 2007), particularly when allowing for possible loss of Ni from the olivine due to reaction with the sulfide liquids (Barnes et al., 2013). Instead, Ni contents of the Xiarihamu olivines are comparable with those of the olivine phenocrysts in picrites and basalts derived from partial melting of pyroxenite in the upper mantle, which is generated by replacement of olivine by pyroxene due to reaction with silica-rich melts (Sobolev et al., 2007). Such magmas have anomalously high Ni, by a factor of 1.5 to 2, relative to normal tholeiite of similar Mg numbers. This suggests that Ni enrichment of the parent magma of the Xiarihamu intrusion may be originally generated from partial melting of a pyroxenite mantle source. This implies that the parental magma of the Xiarihamu intrusion contained more than 300 ppm Ni, in comparison with continental tholeiites and ocean-island basalts that commonly contain 100 to 200 ppm Ni (Barnes and Picard, 1993; Fryer and Greenough, 1995).

Pyroxenite mantle xenoliths in orogenic belts are considered to be converted from lherzolites, through replacement of olivine by pyroxene, owing to reaction with subduction-related silica-rich transient melt. In the north Kamchatka arc in the northwestern Pacific and the late Paleozoic Variscan orogenic belt in Europe, up to 30% of the mantle xenoliths in the basalts are pyroxenites (Kepezhinskas et al., 1995; Ackerman et al., 2009). This makes it possible to produce Ni-rich basaltic magma by moderate-degree partial melting of the pyroxenite mantle in orogenic belts. Thus, it is a reasonable inference that the Ni-rich primary magma of the Xiarihamu intrusion was produced by partial melting of a pyroxenite mantle source in the Middle Kunlun zone.

Both low-degree partial melting of the upper mantle and prior sulfide removal can lead to PGE-depleted parental magma. The PGE depletion of the Ni-Cu sulfide deposits at the southern margins of the Central Asian orogenic belt has been attributed to secondary sulfide segregation or sulfides retaining in the upper mantle source during partial melting (Song and Li, 2009; Li et al., 2012). The extremely low PGE of the Xiarihamu sulfides (Table 3) suggests that the sulfides were segregated from parental magmas extremely depleted in PGE due to low-degree partial melting and retention of PGE in residual sulfides. On the other hand, we still cannot rule out the possibility that the PGE-depleted sulfides at Xiarihamu were segregated from magmas depleted in PGE by prior sulfide extraction from originally PGE-undepleted parental magma.

Sulfides segregated from diverse parental magmas: As described above, the western and eastern parts of the orebody M1 are distinguishable by sulfide contents and host rocks. Especially, the disseminated sulfides in the western part of M1 have unusually high Ni tenors (Ni_{100} = 13–19.3, except for two samples), evidently higher than those of the disseminated sulfides in the middle and eastern parts (Ni_{100} = 3.6–8.2) (Table 2, Figs. 10, 13). The disseminated sulfides plot on two separate trends in the diagrams of Ni versus S (Fig. 9a) and Pd_{100} versus Ir_{100} (Fig. 10d) with exception of the samples mentioned above enriched in Pd due to hydrothermal alteration. The exchange between the sulfide liquids and olivine crystals during the sulfide segregation may have contributed to the high Ni tenors and Ni/Cu ratios of the disseminated ores hosted by the olivine orthopyroxenites (Fig. 7a, b). Serpentinization of olivine is not necessary for the formation of unusually Ni rich sulfides (Barnes et al., 2011).

Reaction between sulfide liquid and olivine is governed by an equilibrium constant (Brenan, 2003):

$$K_D = (X_{NiS}/X_{FeS})^{\text{sulfide liquid}}/(X_{NiO}/X_{FeO})^{\text{olivine}}, \quad (1)$$

where X_i is the mole fraction of the component *i* in the phase (shown at top right corner) of interest. The positive dependence of K_D on the Ni content of the sulfide phase implies that Ni tends to partition more strongly into already Ni-enriched sulfide. This mechanism was considered to have contributed to the high Ni tenors (15–25 wt %) of the disseminated sulfides hosted in harzburgite and orthopyroxenite in the Santa Rita deposit (Brazil) and in some komatiitic dunite bodies (Barnes et al., 2011, 2013). This process needs the sulfide liquid to equilibrate with a mixture of silicate melt and olivine, with a high enough olivine content to upgrade the sulfide in Ni but not in Cu and PGE, hence increasing the Ni/Cu ratio. According to estimation by Li et al. (2015) using the equation of Barnes et al. (2013), K_D of equilibrium between sulfide liquids and olivine crystals in the Xiarihamu deposit is about 10 to 13 under an oxygen fugacity about one log unit above the quartz-fayalite-magnetite buffer (QFM +1). Thus, although sulfide droplets dominantly absorb Ni from silicate melt because of very high $D_{Ni}^{Sul/Sil}$ (= ~500–900; Peach et al., 1990), reaction with olivine may have a limited contribution on the high Ni tenors of the disseminated sulfides (commonly <5 modal % sulfides, up to 15%) hosted in the olivine orthopyroxenites, which contain 10 to 40% olivine (Figs. 5, 6). Model calculations indicated that unusually high Ni tenors (15–25 wt %) of the sulfides in the Santa Rita deposit were segregated from a komatiitic basaltic magma containing ~600 ppm Ni under a silicate melt/sulfide liquid ratio of 500 to 1,000 and olivine crystallized from the Ni-rich magma contain 2,300 to 3,000 ppm Ni (Barnes et al., 2013). According to the model calculations of Barnes et al. (2013), 2,000 to 3,000 ppm Ni in the olivine implies that the disseminated sulfides in the olivine orthopyroxenite at Xiarihamu might be segregated under R factors of 200 to 500 if the parental high Mg basaltic magma contains 600 ppm Ni. If this is true, the parental magmas for the olivine orthopyroxenites at Xiarihamu are extremely low in PGE (0.03–0.22 ppb Pt, 0.15–0.16 ppb Pd, 0.01 ppb Ir), very high in Ni (520–620 ppm), and moderate in Cu (57–128 ppm) according to the calculation using the equation proposed by Campbell and Naldrett (1979):

$$C_s = C_l{*}D{*}(R+1)/(R+D), \quad (2)$$

in which C_s and C_l mean concentration of an element in sulfide liquid and silicate melt, respectively; D is partition coefficient of that element between the sulfide liquid and silicate melt; R is the mass ratio between the silicate melt and sulfide liquid. In the calculation, $D_{Ir}^{Sul/Sil}$, $D_{Pd}^{Sul/Sil}$, and $D_{Ni}^{Sul/Sil}$ are assumed to be 200,000, 100,000, and 800, respectively, according to direct experimental measurements by Mungall and Brenan (2014). Such parental magma is clearly depleted in PGE and unusually enriched in Ni.

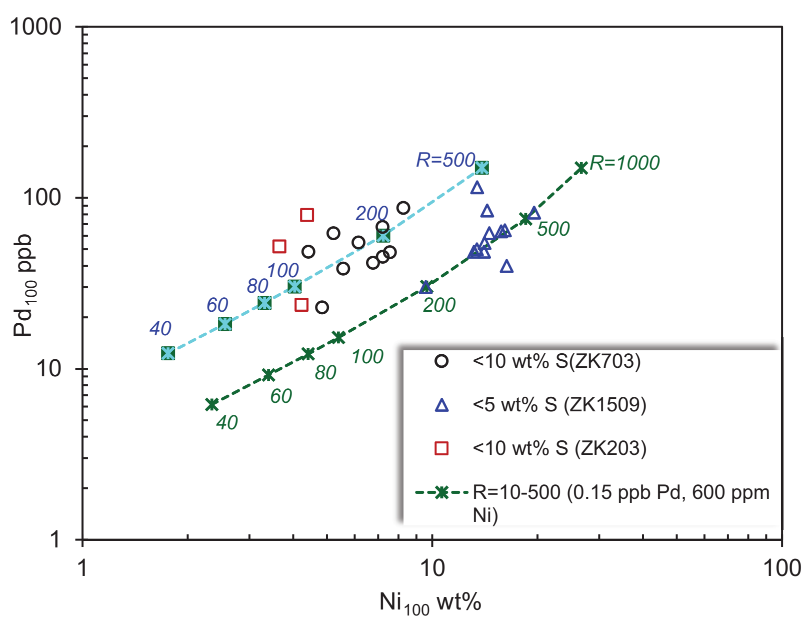

Fig. 13. Model calculation of sulfide segregation shown by diagram of Ni_{100} vs. Pd_{100}, indicating that the sulfides from the western part of the main orebody M1 (ZK1905) were segregated from less evolved magma rich in Ni under higher R factors relative to the sulfides in the middle and eastern part of M1 (ZK703 and ZK203). Reaction with olivine contributes to the increase of Ni tenor in the disseminated sulfides in the western part of M1.

In the binary diagram of Pd_{100} versus Ni_{100} (Fig. 13), most of the disseminated sulfides in the western part of M1 plot along the tendency of the sulfides segregated from a PGE-depleted parental magma containing 0.01 ppb Ir, 0.15 ppb Pd, and 600 ppm Ni under R factors ranging from 200 to 500 (Fig. 13); the two samples with high Pd_{100} may have been modified by hydrothermal alteration as discussed above. In contrast, the disseminated sulfides hosted by the orthopyroxenites in the middle and eastern parts of M1 have relatively high Pd_{100} and low Ni_{100}, implying that their parental magma contains slightly higher Pd (0.3 ppb) and lower Ni (450 ppm) and the sulfides were segregated under lower R values (100–200; Fig. 13). This is supported by the plotting of Pd_{100} versus Ir_{100} (Fig. 12a). The model calculations suggest that the sulfides of the western part of M1 were segregated from less fractionated parental magma containing slightly higher Ni and lower Pd concentrations than those of the middle and eastern parts. This is consistent with the observation that the disseminated sulfides in the western part are hosted in the olivine orthopyroxenite and the sulfide ores in the middle and eastern parts are hosted in orthopyroxenite.

Fractionation of monosulfide solid solution (mss) can result in decrease of Ir and slightly increase of Pd and Pd/Ir ratio in sulfide liquid because Ir is compatible to Mss ($D_{Ir}^{Mss/Sul}$ = 5–17) and Pd is incompatible ($D_{Pd}^{Mss/Sul}$ = 0.13–0.24; Barnes et al., 1997). The small variation in Ir_{100} (1.4–2.0 ppb) of the massive and net-textured ores containing >10% sulfur suggests a weak fractionation of the sulfide melt within the Xiarihamu intrusion (Fig. 12a). This implies that the relatively large variation in Pd_{100} (13.6–79.2 ppb) and large range of Pd/Ir and Ni/Cu ratios (14–32 and 5–68, respectively) of the massive and net-textured ores are associated with removal of Pd and Cu during hydrothermal alteration (Fig. 12).

A Genetic Hypothesis for the Xiarihamu Deposit and Conclusions

The Xiarihamu intrusion, situated in the eastern Kunlun orogenic belt, was most likely formed in a syn- or postcollisional tectonic setting in the Early Devonian. Reaction with silica-rich melt derived from melting of subduction slab converted lherzolite to pyroxenite locally in the mantle wedge before collision. The Xiarihamu intrusion is genetically associated with high Mg basaltic magma derived from partial melting of a pyroxenite mantle source leaving remnant sulfides in the mantle restite. Consequently, the primary magma was S saturated, low in PGE and Cu and high in Ni and Ni/Cu ratio due to a high proportion of pyroxene to olivine in the mantle source.

When the primary magma rose through the crust, it became S undersaturated due to decrease of pressure. Sulfide segregation was triggered by subsequent crustal contamination and sulfur addition, which is indicated by less depleted $\varepsilon_{Hf(t)}$ values of zircon crystals from the Xiarihamu ultramafic rocks (1–5) and positive $\delta^{34}S$ values of sulfide ores (3.5–6.8‰; Li et al., 2015). The sulfide liquid was originally characterized by low PGE and Cu and relatively high in Ni and Ni/Cu ratio. The disseminated sulfides hosted in the olivine orthopyroxenite were segregated from less evolved magma under relatively high R factors (~200–500) and thus are richer in Ir and Ni than those hosted in the orthopyroxenite according to the model calculations shown in Figures12a, and 13. In contrast, the sulfides hosted in the orthopyroxenites were segregated from more evolved magma under low R values (~100–200). Reaction between sulfide liquids and olivine crystals contributed to the higher Ni tenors and Ni/Cu ratios of the disseminated sulfides hosted in the olivine orthopyroxenite than those hosted in the orthopyroxenite.

The geologic features of the orebody M1 were controlled by the geometry of the magma chamber, which constrained the direction of the slurry flow. In the early stage, the ore pod hosted in both olivine orthopyroxenite and orthopyroxenite was formed when the slurry containing more sulfide droplets deposited from a more evolved magma between lines 17 and 7 (Figs. 3, 4, 6). Massive and net-textured sulfides accumulated in embayments at the base. The blotchy textured ores may have formed when the sulfide droplets coalesced and percolated through the ultramafic cumulus pile, or when the settled sulfides were stirred turbulently by a new pulse of magma and transported a small distance. To the east of the exploration line 7, the orebody M1 splits into subore layers toward the southeast, indicating less and less sulfides settling from the magma away from the entrance to the wider portion of the magma conduit. The smaller ore pod hosted in the olivine orthopyroxenite to the west of exploration line 17 was formed by the sulfide droplets settled along with olivine crystals from another pulse of less evolved magma containing sulfides with a higher Ni/Cu ratio.

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REFERENCES

Ackerman, L., Jelínek, E., Medaris, G., Jr., Ježek, J., Siebel, W., and Strnad, L., 2009, Geochemistry of Fe-rich peridotites and associated pyroxenites from Horní Bory, Bohemian massif: Insights into subduction-related melt-rock reactions: Chemical Geology, v. 259, p. 152–167.

Akmaz, R.M., Uysal, I., and Saka, S., 2014, Compositional variations of chromite and solid inclusions in ophiolitic chromites from the southeastern Turkey: Implication for chromite genesis: Ore Geology Reviews, v. 58, p. 208–224.

Arndt, N.T., Lesher, C.M., and Czamanske, G.K., 2005, Mantle-derived magmas and magmatic Ni-Cu-(PGE) deposits: Economic Geology 100th Anniversary Volume, p. 5–24.

Arndt, N.T., Lesher, C.M., and Barnes, S.J., 2008, Komatiite: Cambridge, Cambridge University Press, 467 p.

Barnes, Sarah-J., and Picard, C.P., 1993, The behavior of platinum-group elements during partial melting, fractional crystallization, and sulfide segregation: An example from the Cape Smith fold belt, northern Quebec: Geochimica et Cosmochimica Acta, v. 57, p. 79–87.

Barnes, Sarah-J., Makovicky, E., Makovicky, M., RoseHansen, J., and Karup Moller, S., 1997, Partition coefficients for Ni, Cu, Pd, Pt, Rh and Ir between monosulphide solid solution and sulphide liquid and the implications for the formation of compositionally zoned Ni-Cu sulphide bodies by fractional crystallization of sulphide liquid: Canadian Journal of Earth Sciences, v. 34, p. 366–374.

Barnes, Sarah-J., Cox, R.A., and Zientek, M.L., 2006, Platinum-group element, gold, silver and base metal distribution in compositionally zoned sulfide droplets from the Medvezky Creek mine, Noril'sk, Russia: Contributions to Mineralogy and Petrology, v. 152, p. 187–200.

Barnes, Stephen J., 1993, Partitioning of the platinum group element and gold between silicate and sulphide magmas in the Munni Munni Complex, Australia: Geochimica et Cosmochimica Acta, v. 57, p. 1277–1290.

Barnes, Sarah-J., Coats, C.J.A., and Naldrett, A.J., 1982, Petrogenesis of a Proterozoic nickel-sulfide-komatiite association: the Katiniq sill, Ungava, Quebec: Economic Geololgy, v. 77, p. 413–429.

Barnes, Stephen J., and Fiorentini, M.L., 2012, Komatiite magmas and nickel sulfide deposits: A comparison of variably endowed Archean terranes: Economic Geololgy, v. 107, p. 755–780.

Barnes, Stephen J., and Liu, W., 2012, Pt and Pd mobility in hydrothermal fluids: Evidence from komatiites and from thermodynamic modelling: Ore Geology Reviews, v. 44, p. 49–58.

Barnes, Stephen J., Lesher, C.M., and Keays, R.R., 1995, Geochemistry of mineralized and barren komatiites from the Perseverance nickel deposit, Western Australia: Lithos, v. 34, p. 209–234.

Barnes, Stephen J., Osborne, G.A., Cook, D., Barnes, L., Maier, W.D., and Godel, B., 2011, The Santa Rita nickel sulfide deposit in the Fazenda Mirabela intrusion, Bahia, Brazil: Geology, sulfide geochemistry, and genesis: Economic Geology, v. 106, p. 1083–1110.

Barnes, Stephen J., Godel, B., Gürer, D., Brenan, J.M., Robertson, J., and Paterson, D., 2013, Sulfide-olivine Fe-Ni exchange and the origin of anomalously Ni rich magmatic sulfides: Economic Geololgy, v. 108, p. 1971–1982.

Beresford, S., Stone, W.E., Cas, R., Lahaye, Y., and Jane, M., 2005, Volcanological controls on the localization of the komatiite-hosted Ni-Cu-(PGE) Coronet deposit, Kambalda, Western Australia: Economic Geololgy, v. 100, p. 1457–1467.

Bian, Q.T., Li, D.H., Pospelov, I., Yin, L.M., Li, H.S., Zhao, D.S., Chang, C.F., Luo, X.Q., Gao, S.L., Astrakhantsevc, O., and Chamov, N., 2004, Age, geochemistry and tectonic setting of Buqingshan ophiolites, North Qinghai-Tibet plateau, China: Journal of Asian Earth Sciences, v. 23, p. 577–596.

Black, L.P., Kamo, S.L., Allen, C.M., Aleinikoff, J.N., Davis, D.S., Korsch, R.J., and Foudoulis, C., 2003, TEMORA 1: A new zircon standard for Phanerozoic U-Pb geochronology: Chemical Geology, v. 200, p. 155–170.

Brenan, J.M., 2003, Effects of of f_{O_2}, f_{S_2}, temperature, and melt composition on Fe-Ni exchange between olivine and sulfide liquid: Implications for natural olivine-sulfide assemblages: Geochimica et Cosmochimica Acta, v. 67, p. 2663–2681.

Campbell, I.H., and Naldrett, A.J., 1979, The influence of silicate:sulfide ratios on the geochemistry of magmatic sulfides: Economic Geology, v. 74, p. 1503–1506.

Chen, L.-M., Song, X.-Y., Keays, R.R., Tian, Y.-L., Wang, Y.-S., Deng, Y.-F., and Xiao, J.-F., 2013, Segregation and fractionation of magmatic Ni-Cu-PGE sulfides in the western Jinchuan intrusion, northwestern China: Insights from platinum group element geochemistry: Economic Geology, v. 108, p. 1793–1811.

Chen, L.-M., Song, X.-Y., Danyushevsky, L.V., Wang, Y.-S., Tian, Y.-L., and Xiao, J.-F., 2015, A laser ablation ICP-MS study of platinum-group and chalcophile elements in base metal sulfide minerals of the Jinchuan Ni-Cu sulfide deposit, NW China: Ore Geology Reviews, v. 65, p. 955–967.

Chen, N.S., Sun, M., He, L., Zhang, K.X., and Wang, G.C., 2002, Precise timing of the early Paleozoic metamorphism and thrust deformation in the eastern Kunlun orogen: Chinese Science Bulletin, v. 46, p. 330–333 (in Chinese with English ab.).

Chen, N.S., Li, X.Y., Wang, X.Y., Chen, Q., Wang, Q.Y., and Wan, Y.S., 2006, Zircon SHRIMP U-Pb age of Neoproterozoic metagranite in the North Kunlun unit on the southern margin of the Qaidam block in China: Geological Bulletin of China, v. 25, p. 1311–1314 (in Chinese with English abs.).

Chung, H.-Y., and Mungall, J.E., 2009, Physical constraints on the migration of immiscible fluids through partially molten silicates, with special reference to magmatic sulfide ores: Earth and Planetary Science Letters, v. 286, p. 14–22.

Compston, W., Williams, I.S., and Kirschvink, J.L., 1992, Zircon U-Pb ages for the early Cambrian timescale: Journal of the Geological Society, v. 149, p. 171–184.

Crawford, A.J., Falloon, T.J., and Green, T.H., 1989, Classification, petrogenesis and tectonic setting of boninites, *in* Crawford, A.J., ed., Boninites and related rocks: Unwin Hyman, p. 1–49.

Cui, M.H., Meng, F.C., and Wu, X.K., 2011, Early Ordovician island arc of Yaziquan, west of Qimantag Mountain, eastern Kunlun: Evidences from geochemistry, Sm-Nd isotope and geochronology of intermediate-basic igneous rocks: Acta Petrologica Sinica, v. 27, p. 3365–3379 (in Chinese with English abs.).

Deng, Y.-F., Song, X.-Y., Chen, L.-M., Zhou, T.-F., Pirajno, F., and Yuan, F., 2014, Geochemistry of the Huangshandong Ni-Cu deposit in northwestern China: Implications for the formation of magmatic sulfide mineralization in orogenic belts: Ore Geology Reviews, v. 56, p. 181–198.

Eby, G.N., 1992, Chemical subdivision of the A-type granitoids: Petrogenetic and tectonic implications: Geology, v. 20, p. 641–644.

Evans-Lamswood, D.M., Butt, D.P., Jackson, R.S., Lee, D.V., Muggeridge, M.G., Wheeler, R.I., and Wilton, D.H.C., 2000, Physical controls associated with the distribution of sulfides in the Voisey's Bay Ni-Cu-Co deposit, Labrador: Economic Geology, v. 95, p. 749–769.

Fleet, E.M., and Stone, W.E., 1991, Partitioning of platinum-group elements in the Fe-Ni-S system and their fractionation in nature: Geochimica et Cosmochimica Acta, v. 55, p. 245–253.

Fryer, B.J., and Greenough, J.D., 1995, Evidence for mantle heterogeneity from platinum-group element abundances in Indian Ocean basalts: Canadian Journal of Earth Sciences, v. 29, p. 2329–2339.

Gammons, C.H., Bloom, M.S., and Yu, Y., 1992, Experimental investigations of the hydrothermal geochemistry of platinum and palladium. I. Solubility of platinum and palladium sulphide minerals in NaCl/H_2SO_4 solutions at 300°C: Geochimica et Cosmochimica Acta, v. 56, p. 3881–3894.

Gao, J.F., Zhou, M.F., Lightfoot, P.C., Wang, C.Y., Qi, L., and Sun, M., 2013, Sulfide saturation and magma emplacement in the formation of the Permian Huangshandong Ni-Cu sulfide deposit, Xinjiang, northwestern China: Economic Geology, v. 108, p. 1833–1848.

Gao, Y.B., and Li W.Y., 2011, Petrogenesis of granites containing tungsten and tin ores in the Baiganhu deposit, Qimantage, NW China: Constraints from petrology, chronology and geochemistry: Geochimica, v. 40, p. 324–336 (in Chinese with English abs.).

Gao, Y.B., Li, W.Y., Xie, X., Zhang, Z.W., Guo Z.P., Zhang, J.W., and Li, K.K., 2012, Geology, geochemistry and genesis of the Lashuixia Ni-Cu deposit in Hualong area, Qinghai province: Geological Bulletin of China, v. 31, p. 763–777 (in Chinese with English abs.).

Huang, H., Niu, Y.L., Nowell, G., Zhao, Z.D., Yu, X.H., Zhu, D.C., Mo, X.X., and Ding, S., 2014, Geochemical constraints on the petrogenesis of granitoids in the East Kunlun orogenic belt, northern Tibetan plateau: Implications for continental crust growth through syn-collisional felsic magmatism: Chemical Geology, v. 370, p. 1–18.

Jiang, C.F., Yang, J.S., Feng, B.G., Zhu, Z.Z., Zhao, M., Chai, Y.C., Shi, X.D., Wang, H.D., and Hu, J.Q., 1992, Opening-closing tectonics of Kunlun Mountains: Beijing, Geological Publishing House, 224 p. (in Chinese with English abs.).

Keays, R.R., 1995, The role of komatiitic and picritic magmatism and S-saturation in the formation of ore deposits: Lithos, v. 34, p. 1–18.

Keays, R.R., Ross, J.R., and Woolrich, P., 1981, Precious metals in volcanic peridotite-associated nickel sulfide deposits in Western Australia. II. Distribution within the ores and host rocks at Kambalda: Economic Geology, v. 76, p. 1645–1674.

Kepezhinskas, P.K., Defant, M.J., and Drummond, M.S., 1995, Na metasomatism in the island-arc mantle by slab melt-peridotite interaction: Evidence from mantle xenoliths in the north Kamchatka arc: Journal of Petrology, v. 36, p. 1505–1527.

Kerr, A., and Leitch, A.M., 2005, Self-destructive sulfide segregation systems and the formation of high-grade magmatic ore deposits: Economic Geology, v. 100, p. 311–332.

Lesher, C.M., and Campbell, I.H., 1993, Geochemical and fluid dynamic modeling of compositional variations in Archean komatiite-hosted nickel sulfide ores in Western Australia: Economic Geology, v. 88, p. 804–816.

Lesher, C.M., and Keays, R.R., 2002, Komatiite-associated Ni-Cu-(PGE) deposits: Canadian Institute of Mining, Metallurgy and Petroleum Special Volume 54, p. 579–617.

Li, C., and Ripley, E.M., 2009, Sulfur contents at sulfide-liquid or anhydrite saturation in silicate melts: Empirical equations and example applications: Economic Geology, v. 104, p. 405–412.

Li, C., Naldrett, A.J., and Ripley, E.M., 2001, Critical factors for the formation of a nickel-copper deposit in an evolved magma system: Lessons from a comparison of the Pants Lake and Voisey's Bay sulfide occurrences in Labrador, Canada: Mineralium Deposita, v. 36, p. 85–92.

Li, C., Zhang, M., Fu, P., Qian, Z., Hu, P., and Ripley, E.M., 2012, The Kalatongke magmatic Ni-Cu deposits in the Central Asian orogenic belt, NW China: Product of slab window magmatism?: Mineralium Deposita, v. 47, p. 51–67.

Li, C., Zhang, Z., Li, W., Wang, Y., Sun, T., and Ripley, E.M., 2015, Geochronology, petrology and Hf-S isotope geochemistry of the newly discovered Xiarihamu magmatic Ni-Cu sulfide deposit in the Qinghai-Tibet plateau, western China: Lithos, v. 216–217, p. 224–240.

Lightfoot, P.C., and Keays, R.R., 2005, Siderophile and chalcophile metal variations in flood basalts from the Siberian Trap, Noril'sk region: Implications for the origin of the Ni-Cu-PGE sulfide ores: Economic Geology, v. 100, p. 439–462.

Lightfoot, P.C., Keays, R.R., Evans-Lamswood, D., and Wheeler, R.S., 2012, Saturation history of Nain Plutonic Suite mafic intrusions: Origin of the Voisey's Bay Ni-Cu-Co sulfide deposit, Labrador, Canada: Mineralium Deposita, v. 47, p. 23–50.

Liu, B., Ma, C.Q., Zhang, J.Y., Xiong, F.H., Huang J., and Jiang H.A., 2012, Petrogenesis of Early Devonian intrusive rocks in the east part of eastern Kunlun orogen and implication for early Palaeozoic orogenic processes: Acta Petrologica Sinica, v. 28, p. 1785–1807 (in Chinese with English abs.).

Liu, B., Ma, C-Q., Guo, P., Zhang, J.-Y., Xioung, F.-H., Huang, J., and Jiang, H.-A., 2013, Discovery of the middle Devonian A-type granite from the eastern Kunlun orogen and its tectonic implication: Earth Science, v. 38(5), p. 947–962.

Liu, W., and McPhail, D.C., 2005, Thermodynamic properties of copper chloride complexes and copper transport in magmatic-hydrothermal solutions: Chemical Geology, v. 221, p. 21–39.

Liu, Y.J., Genser, J., Neubaue, F., Jin, W., Ge, X.H., Handle, R., and Takasu, A., 2005, $^{40}Ar/^{39}Ar$ mineral ages from basement rocks in the eastern Kunlun Mountains, NW China, and their tectonic implications: Tectonophysics, v. 398, p. 199–224.

Lu, L., Wu, Z.H., Hu, D.G., Barosh, P.J., Hao, S., and Zhou, C.J., 2010, Zircon U-Pb ages for rhyolite of the Maoniushan formation and its tectonic significance in the east Kunlun Mountains: Acta Petrologica Sinica, v. 26, p. 1150–1158 (in Chinese with English abs.).

Ludwig, K.R., 2001a. Squid 1.02. A user manual: Berkeley Geochronological Center Special Publication, p. 1–219.

——2001b. Using Isoplot/EX, version 2.49. A Geochronological Toolkit for Microsoft Excel: Berkeley Geochronological Center Special Publication, p. 1–55.

Maier, W.D., Barnes, Sarah-J., Chinyepi, G., Barton, J.M., Eglington, B., and Setshedi, I., 2008, The composition of magmatic Ni-Cu-(PGE) sulfide deposits in the Tati and Selebi-Phikwe belts of eastern Botswana: Mineralium Deposita, v. 43, p. 37–60.

Makkonen, H.V., Mäkinen, J., and Kontoniemi, O., 2008, Geochemical discrimination between barren and mineralized intrusions in the Svecofennian (1.9 Ga) Kotalahti nickel belt, Finland: Ore Geology Reviews, v. 33, p. 101–114.

Mao, J.W., Pirajno, F., Zhang, Z.H., Chai, F.M., Wu, H., Chen, S.P., Chen, L.S., Yang, J.M., and Zhang, C.Q., 2008, A review of the Cu-Ni sulphide deposits in the Chinese Tianshan and Altay orogens (Xinjiang autonomous region, NW China): Principal characteristics and ore-forming processes: Journal of Asian Earth Sciences, v. 32(2–4), p. 184–203.

Mao, Y.J., Qin, K.Z., Li, C.S., Xue, S.C., and Ripley, E.M., 2014, Petrogenesis and ore genesis of the Permian Huangshanxi sulfide ore-bearing mafic-ultramafic intrusion in the Central Asian orogenic belt, western China: Lithos, v. 200–201, p. 111–125.

Meng, F., Zhang, J., and Cui, M., 2013a, Discovery of early Paleozoic eclogite from the east Kunlun, western China and its tectonic significance: Gondwana Research, v. 23, p. 825–836.

Meng, F.C., Cui, M.H., Wu, X.K., Wu, J.F., and Wang, J.H., 2013b, Magmatic and metamorphic events recorded in granitic gneisses from the Qimantag, east Kunlun Mountains, Northwest China: Acta Petrologica Sinica, v. 29, p. 2107–2212 (in Chinese with English abs.).

Mo, X.X., Luo, Z.H., Deng, J.F., Yu, X.H., Liu, C.D., Chen, H.W., Yuan, W.M., and Liu, Y.H., 2007, Granitoids and crustal growth in the east Kunlun orogenic belt: Geological Journal of China Universities, v. 13, p. 403–414 (in Chinese with English abs.).

Mungall, J.E., and Brenan, J.M., 2014, Partitioning of platinum-group elements and Au between sulfide liquid and basalt and the origins of mantle-crust fractionation of the chalcophile element: Geochimica et Cosmochimica Acta, v. 125, p. 265–289.

Naldrett, A.J., 1998, World-class Ni-Cu-PGE deposits: Key factors in their genesis: Mineralium Deposita, v. 34, p. 227–240.

——2004, Magmatic sulfide deposits—geology, geochemistry and exploration: Berlin, Heidelberg, New York, Springer.

Nasdala, L., Hofmeister, W., Norberg, N., Martinson, J.M., Corfu, F., Dörr, W., Kamo, S.L., Kennedy, A.K., Kronz, A., Reiners, P.W., Frei, D., Kosler, J., Wan, Y.S., Götze, J., Hager, T., Kröner, A., and Valley, J.W., 2008, Zircon M257—a homogeneous natural reference material for the ion microprobe U-Pb analysis of zircon: Geostandards and Geoanalytical Research, v. 32, p. 247–265.

No. 5 GMSI (No. 5 Geological and Mineral Survey Institute of Qinghai Bureau of Geology and Mineral Resources), 2013, Exploration design for the Xirihamu Ni-Cu sulfide deposit, Gulmud, Qinghai province, 169 p. (in Chinese).

Pagé, P., and Barnes, Sarah-J., 2009, Using trace elements in chromites to constrain the origin of podiform chromites in the Thetford Mines ophiolite, Québec, Canada: Economic Geology, v. 104, p. 997–1018.

Pan, Y.S., Zhou, W.M., Xu, R.H., Wang, D.A., Zhang, Y.Q., Xie, Y.W., Chen, T.E., and Luo, H., 1996, Geological characteristics and evolution of the Kunlun Mountains region during the early Paleozoic: Science in China (Series D), v. 26, p. 302–307.

Peach, C.L., Mathez, E.A., and Keays, R.R.,1990, Sulfide melt/silicate melt distribution coefficients for noble metals and other chalcophile elements as deduced from MORB: Implications for partial melting: Geochimca et Cosmochimica Acta,v. 54, p. 3379–3389.

Peltonen, P., 1995, Petrogenesis of ultramafic rocks in the Vammala nickel belt: Implications for crustal evolution of the early Proterozoic Svecofennian arc terrane: Lithos, v. 34, p. 253–274.

Piňa, R., Gervilla, F., Ortega, L., and Lunar, R., 2008, Mineralogy and geochemistry of platinum-group elements in the Aguablanca Ni-Cu deposit (SW Spain): Mineralogy and Petrology, v. 92, p. 259–282.

Piňa, R., Romeo, I., Ortega, L., Lunar, R., Capote, R., Gervilla, F., Tejero, R., and Quesada, C., 2010, Origin and emplacement of the Aguablanca magmatic Ni-Cu-(PGE) sulfide deposit, SW Iberia: A multidisciplinary approach: Geological Society of America Bulletin, v. 122, p. 915–925.

Ripley, E.M., Li, C., and Shin, D., 2002, Paragneiss assimilation in the genesis of magmatic Ni-Cu-Co sulfide mineralization at Voisey's Bay, Labrador: $\delta^{34}S$, $\delta^{13}C$, and Se/S evidence: Economic Geology, v. 97, p. 1307–1318.

Qi, L., Zhou M.-F., and Wang C.Y., 2004, Determination of low concentrations of platinum group elements in geological samples by ID-ICP-MS: Journal of Analytical Atomic Spectrometry, v. 19, 1335–1339.

Qi, L., Gao J.-F., Huang X.-W., Hu J., Zhou M.-F., and Zhong H., 2011, An improved digestion technique for determination of platinum group elements in geological samples: Journal of Analytical Atomic Spectrometry, v. 26, p. 1900–1904.

Qin, K.Z., Su, B.X., Sakyi, P.A., Tang, D.M., Li, X.H., Sun, H., Xiao, Q.H. and Liu, P.P., 2011, SIMS zircon U-Pb geochronology and Sr-Nd isotopes of Ni-Cu-bearing mafic-ultramafic intrusions in eastern Tianshan and Beishan in correlation with flood basalts in Tarim basin (NW China): Constraints on a ca. 280 Ma mantle plume: American Journal of Science, v. 311, p. 237–260.

Sobolev, A.V., Hoffman, A., Kuzmin, D., Yaxley, G., Arndt, N., Chung,S.-L., Danyushevsky, L., Elliott, T., Frey, F., Garcia, M., Gurenko, A.,Kamenetsky, V., Kerr, A., Krivolutskaya, N., Matvienkov, V., Nikogosian, I.,Rocholl, A., Sigurdson, I., Sushchevskaya, N., and Teklay, M., 2007, The amount of recycled crust in sources of mantle-derived melts: Science, v. 316, p. 412–417.

Song, B., Zhang, Y.H., Wan, Y.S., and Jian, P., 2002, The SHRIMP sample manufacture, test and explanation of some phenomena for the zircon: Geological Review, v. 48, p. 26–30 (in Chinese with English abs.).

Song, X.-Y. and Li, X.-R., 2009, Geochemistry of the Kalatongke Ni-Cu-(PGE) sulfide deposit, NW China: Implications for the formation of magmatic sulfide mineralization in a post-collisional environment: Mineralium Deposita, v. 44, p. 303–327.

Song, X.Y., Zhou, M.F., Cao, Z.M., Sun, M., and Wang, Y.L., 2003, Ni-Cu-(PGE) magmatic sulfide deposits in the Yangliuping area, Permian Emeishan igneous province, SW China: Mineralium Deposita, v. 38, p. 831–843.

Song, X.-Y., Zhou, M.-F., Wang, Y.C., Qi, L., and Zhang, C.-J., 2006, Role of crustal contamination in formation of the Jinchuan intrusion and its world-class Ni-Cu-(PGE) sulfide deposit, northwestern China: International Geology Review, v. 48, p. 1113–1133.

Song, X.-Y., Zhou, M.-F., Tao, Y., and Xiao, J.-F., 2008, Controls on the metal compositions of magmatic sulfide deposits in the Emeishan large igneous province, SW China: Chemical Geology, v. 253, p. 38–49.

Song, X.-Y., Keays, R.R., Zhou, M.-F., Qi, L., Ihlenfeld, C., and Xiao, J.-F., 2009a, Siderophile and chalcophile elemental constraints on the origin of the Jinchuan Ni-Cu-(PGE) sulfide deposit, NW China: Geochimica et Cosmochimica Acta, v. 73, p. 404–424.

Song, X.-Y., Hu, R.-Z., and Chen, L.-M., 2009b, Geochemical natures of copper, nickel and PGE and their significance for the study of origin and evolution of mantle-derived magmas and magmatic sulfide deposits: Earth Science Frontiers, v. 16, p. 287–305.

Song, X.Y., Xie, W., Deng, Y.F., Crawford, A.J., Zheng, W.Q., Zhou, G.F., Deng, G., Cheng, S.L. and Li, J., 2011, Slab break-off and the formation of Permian mafic-ultramafic intrusions in southern margin of Central Asian orogenic belt, Xinjiang, NW China: Lithos, v. 127, p. 128–143.

Song, X.-Y., Chen, L.-M., Deng, Y.-F., and Xie, W., 2013, Syn-collisional tholeiitic magmatism induced by asthenosphere upwelling due to slab detachment at the southern margin of the Central Asian orogenic belt: Journal of the Geological Society, London, v. 170, p. 941–950.

Stacey, J.S., and Kramers, J.D., 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Steiger, R.H., and Jager, E., 1977, Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology: Earth and Planetary Science Letters, v. 36, p. 359–362.

Su, S.G., Li, C., Zhou, M.-F., Ripley, E.M., and Qi, L., 2008, Controls on variations of platinum-group element concentrations in the sulfide ores of the Jinchuan Ni-Cu deposit, western China: Mineralium Deposita, v. 43, p. 609–622.

Wang, G., Sun, F.Y., Li, B.L., Li, S.J., Zhao, J.W., Yang, Q.A., and Ao, Z., 2013, Zircon U-Pb geochronology and geochemistry of the Early Devonian syenogranite in the Xiarihamu ore district from east Kunlun, with implications for the geodynamic setting: Geotectonica et Metallogenia, v. 139, p. 685–697 (in Chinese with English abs.).

Wei, B., Yan Wang, C.W., Li, C., and Sun, Y., 2013, Origin of PGE-depleted Ni-Cu sulfide mineralization in the Triassic Hongqiling No. 7 orthopyroxenite intrusion, Central Asian orogenic belt, northeastern China: Economic Geology, v. 108, p. 1813–1831.

Williams, I.S., 1998, U-Th-Pb geochronology by ion microprobe: Reviews in Economic Geology v. 7, p. 1–35.

Wood, S.A., 2002, The aqueous geochemistry of the platinum-group elements with applications to ore deposits: Canadian Institute of Mining and Metallurgy Special Volume 54, p. 211–249.

Xie, W., Song, X.-Y., Deng, Y.-F., Wang, Y.-S., and Ba, D.-H., 2012, Geochemistry and petrogenetic implications of a Late Devonian mafic-ultramafic intrusion at the southern margin of the Central Asian orogenic belt: Lithos, v. 144–145, p. 209–230.

Xie, W., Song, X.-Y., Chen, L.-M., Deng, Y.-F., Zheng, W.-Q., Wang, Y.-S., Ba, D.-H., Yin, M-H., and Luan, Y., 2014, Geochemistry insights on the genesis of the subduction-related heishan magmatic Ni-Cu-(PGE) deposit in Gansu, northwestern China, at the southern margin of the Central Asian orogenic belt: Economic Geology, v. 109, p. 1563–1583.

Xiong, F., Ma, C., Jiang, H., Liu, B., and Huang, J., 2014, Geochronology and geochemistry of Middle Devonian mafic dykes inthe East Kunlun orogenic belt, northern Tibet plateau: Implications for the transition from Prototethys to Paleotethys orogeny: Chemie der Erde, v. 74, p. 225–235.

Xiong, Y., and Wood, S.A., 2000, Experimental quantification of hydrothermal solubility of platinum-group elements with special reference to porphyry copper environments: Mineralogy and Petrology, v. 68, p. 1–28.

Xu, Z.Q., Yang, J.S., Jiang, M., Li, H.B., Xue, G.Q., Yuan, X.C., and Qian, H., 2001, Deep structure and lithospheric shear faults in the East Kunlun-Qiangtang region, northern Tibetan plateau: Science in China Series D-Earth Sciences, v. 44, p. 1–9.

Xu, Z.Q., Yang, J.S., Li, H.B., Zhang, J.X., and Wu, C.L., 2007, Orogenic plateau: Terrane amalgamation, collision and uplift in the Qinghai–Tibet plateau: Beijing, Geological Publishing House, p. 1–458 (in Chinese).

Yang, J.S., Robinson, P.T., Jiang, C.F., and Xu, Z.Q., 1996, Ophiolites of the Kunlun Mountains, China and their tectonic implications: Tectonophysics, v. 258, p. 215–231.

Yang, J.Z., Shen, Y.C., Li, G.M., Liu, T.B., and Zeng, Q.D., 1999, Basic features and its tectonic significance of Yaziquan ophiolite belt in eastern Kunlun orogenic belt, Xinjiang: Geoscience Journal of China University of Geosciences, v. 13, p. 309–314 (in Chinese with English abs.).

Yang, J.Z., Shen, Y.C., and Liu, T.B., 2000, Tectonic environment analysis on volcanic rocks from Qimantage group, east Kunlun orogenic belt, Xinjiang: Xinjiang Geology, v. 18, p. 105–112 (in Chinese with English abs.).

Zhang, M.J., Li, C., Fu, P.E., Hu, P.Q., and Ripley, E.M., 2011, The Permian Huangshanxi Cu-Ni deposit in western China: Intrusive-extrusive association, ore genesis, and exploration implications: Mineralium Deposita, v. 46, p. 153–170.

Zhang, Z.W., Li, W.Y., Gao, Y.B., Xie, X., Wang, Y.L., Zhang, J.W., Guo, Z.P., and Li, K., 2012, Geochemical characteristics of the Lashuixia basic complex in Qinghai province and its constraints on genesis of the deposit: Geology and Exploration, v. 48, p. 959–968.

Zhou, M.-F., Sun, M., Keays, R.R., and Kerrich, R.W., 1998, Controls on platinum-group elemental distributions: A case study of high-Cr and high-Al chromites from Chinese orogenic belts: Geochimica et Cosmochimica Acta, v. 62, p. 677–688.